
Sympatheia: Emotionally Adaptive Voice Assistant with Continuous Affect Conditioning

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Empathetic spoken dialogue systems must infer a user’s emotional state to respond
2 appropriately, yet everyday speech often carries weak, neutral, or ambiguous
3 affective cues. To address this, we introduce SYMPATHEIA, a speech-to-speech
4 dialogue framework conditioned on affect inferred from the user’s speech and, when
5 available, explicit affect specifications provided as a continuous valence–arousal
6 (VA) control signal by a multimodal sensing module or user interface. To train our
7 model, we construct SYMPATHEIA-18k, an emotion-conditioned synthetic spoken
8 dialogue corpus with 12 emotion anchors. This dataset includes an emotional split
9 for learning affective speech behavior, and a neutral split that pairs emotionally
10 neutral queries with multiple emotion-conditioned responses to isolate explicit
11 emotion control in emotionally ambiguous cases. Empirical results show that
12 SYMPATHEIA outperforms speech conversational baselines in generating responses
13 whose semantic content and spoken delivery are both emotionally appropriate.
14 We further show that the same VA interface can integrate emotion estimates from
15 diverse sensing modules, including facial expression, biosignals, and textual affect
16 descriptions, improving response alignment when speech alone provides limited
17 emotional evidence. These results suggest that continuous affect conditioning is an
18 effective practical step for building emotionally adaptive voice assistants.

19 1 Introduction

20 Voice-based conversational agents are becoming increasingly capable of flexible and open-ended
21 conversational interaction, driven by recent advances in large language models (LLMs) [1, 2]. As
22 these systems evolve from task-oriented tools to companions for extended interaction, response
23 quality depends not only on factual correctness but also on interpersonal sensitivity. Emotional
24 alignment plays an important role in effective communication, as it can make the assistant feel more
25 natural and human-like, reduce misunderstanding, and support interactions that feel more comforting
26 and engaging. In contrast, affectively mismatched responses can alter how a message is perceived
27 and weaken the overall quality of the interaction. Figure 1 illustrates why this capability is essential
28 for an empathetic spoken companion.

29 Prior work on empathetic and emotion-aware dialogue has made important progress in generating
30 more contextually appropriate, supportive, and affect-sensitive responses [3], but two key gaps remain.
31 First, many methods are optimized for spoken inputs with explicit, high-intensity emotional cues.
32 In practice, however, a central challenge is that everyday speech often contains subtle, ambiguous,
33 or mixed affective states that may not be reliably inferred from lexical or acoustic cues alone [4, 5].
34 Human listeners naturally resolve this ambiguity by integrating multiple cues and adapting their
35 responses empathetically. In contrast, current spoken dialogue systems often have limited mechanisms
36 for combining ambiguous emotional cues in user’s speech with external emotional cues beyond speech.
37 As a result, they may generate semantically appropriate yet emotionally misaligned responses.

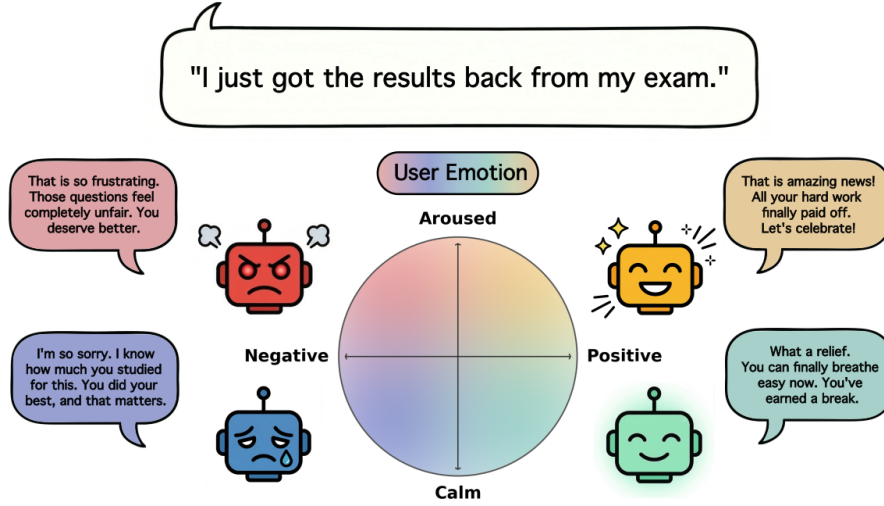


Figure 1: Motivation for SYMPATHEIA. The same user utterance may call for different responses depending on the speaker’s emotional state, illustrated in a valence-arousal space. The system aims to generate spoken responses that are both semantically appropriate and emotionally aligned.

38 Second, emotion is often represented using a fixed set of discrete categories, which can be overly
 39 restrictive for natural interaction. Evidence from affective computing suggests that real-world
 40 emotional expression varies across speakers and cultures, and that continuous representations such as
 41 valence-arousal can better capture this nuance [6]. However, many widely used emotion resources
 42 are still organized around a small set of discrete emotion categories, which can limit how well they
 43 capture the variability and subtlety of naturalistic speech [7, 8].

44 To address these limitations, we introduce SYMPATHEIA, an emotion-conditioned voice dialogue
 45 model for adaptive empathetic response generation. SYMPATHEIA combines (i) implicit affect
 46 inference from user speech with (ii) optional explicit affect control through a continuous valence-
 47 arousal (VA) interface. During training, the 12 named emotions serve as interpretable anchor points
 48 in the VA plane. Each anchor is encoded by a valence-arousal coordinate, allowing the model to learn
 49 affect control in a continuous space rather than as a closed set of class labels. At inference time, this
 50 same interface can accept discrete labels mapped to anchors, continuous VA coordinates, or estimates
 51 from external affect modules such as biosignals, facial expressions, and textual affect descriptions.¹

52 Our main contributions are: (1) We propose SYMPATHEIA, an empathetic voice chatbot framework
 53 conditioned on both speech-inferred affect and explicit valence-arousal cues from emotion sensing
 54 modules or a user interface. Across different evaluation settings, SYMPATHEIA outperforms strong
 55 spoken-dialogue baselines in empathy-oriented response quality. (2) We introduce the SYMPATHEIA-
 56 18k dataset with 18k emotion-conditioned synthetic spoken query-response pairs, including both
 57 emotionally expressive and neutral interactions. (3) We present a modular interface for incorporating
 58 external affect sources, including facial expressions, biosignals such as EEG, ECG or eye tracking,
 59 and textual affect descriptions, enabling multimodal emotion grounding in real-world deployments.

60 2 Related Work

61 **Text-based empathetic dialogue.** Empathetic dialogue research has been driven largely by text-
 62 based benchmarks and generators. EmpatheticDialogues established the standard benchmark for
 63 supportive open-domain response generation [3], and subsequent models such as MoEL and MIME
 64 improved text-based empathy by explicitly routing generation through inferred listener types or
 65 emotion mixtures [9, 10]. These systems established the task formulation most later work inherits,
 66 but they primarily reason over lexical context and discrete emotion supervision rather than prosody
 67 or other paralinguistic cues.

¹Code and audio demos are anonymously available at <https://anonymous.4open.science/r/sympatheia-1181>.

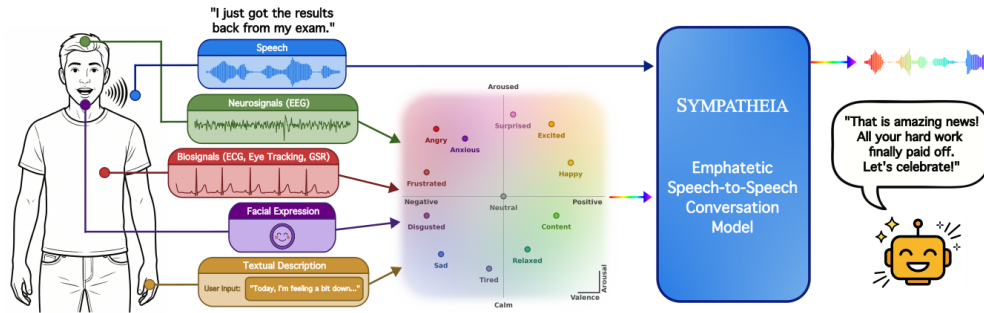


Figure 2: System overview of SYMPATHEIA. The core speech-to-speech dialogue model uses affective cues in the user’s spoken query directly and can additionally receive explicit valence–arousal conditioning from pluggable emotion sensing modules, such as facial expression, EEG/physiological signals, text descriptions, or direct interface selection. These optional external estimates are converted into a shared VA interface that guides emotionally aligned spoken response generation.

Emotion datasets and affect representation. Frequently used emotion datasets span both categorical and continuous annotation schemes. IEMOCAP and MELD have supported emotion recognition in dialogue with discrete labels and multimodal context [7, 8], while resources such as MSP-Podcast, AffectNet, and DEAP include dimensional affect annotations such as valence and arousal across speech, face, and physiological modalities [11–14]. This line of work motivates valence–arousal as a compact interface for external sensing modules, because it can represent graded, mixed, and speaker-dependent affect more flexibly than a small fixed taxonomy [6].

Speech-native dialogue models. Recent spoken-dialogue research has moved beyond cascaded automatic speech recognition (ASR) → LLM → text-to-speech (TTS) pipelines toward models that consume and/or generate speech more directly. Early speech-language systems such as SpeechGPT explored cross-modal conversational modeling [15], and newer systems including Moshi, GLM-4-Voice, Qwen3-Omni, and Kimi-Audio further emphasize real-time interaction and preservation of timing or other paralinguistic cues [16–19]. These models advance speech-native interaction, but they do not primarily formulate empathy as a controllable response-generation objective.

Empathetic spoken dialogue. Closest to our setting are recent empathetic speech-language systems. BLSP-Emo aligns spoken inputs with empathetic text continuations, while OpenS2S, Empathy Omni, and OSUM-EChat extend empathetic dialogue to speech response generation [20–23]. These systems primarily learn affective or paralinguistic context from the user’s spoken query itself. SYMPATHEIA preserves this speech-based affect inference, but additionally exposes emotion as an optional continuous valence–arousal control signal, allowing external sensors or user controls to guide response generation when speech alone is neutral, subtle, or ambiguous.

3 Methodology

3.1 System overview

SYMPATHEIA is an empathetic speech-to-speech conversational model, designed to generate spoken responses that are both semantically appropriate and emotionally aligned. Figure 2 shows the system overview of SYMPATHEIA. While the model already infers user affect from the semantic and acoustic cues present in the spoken query, it also supports system or user-provided external affect information when available. In the broader deployment setting, the overall SYMPATHEIA system combines this core dialogue model with optional pluggable emotion sensing modules, yielding two components: (i) the SYMPATHEIA speech-to-speech model itself and (ii) upstream emotion recognition modules that provide additional affect estimates from other available modalities, such as facial expressions, biosignals such as EEG and ECG, textual affect descriptions or affect selection in user interface. These external signals can provide useful complementary information when the user’s emotional state is subtle, ambiguous, or only partially expressed in speech.

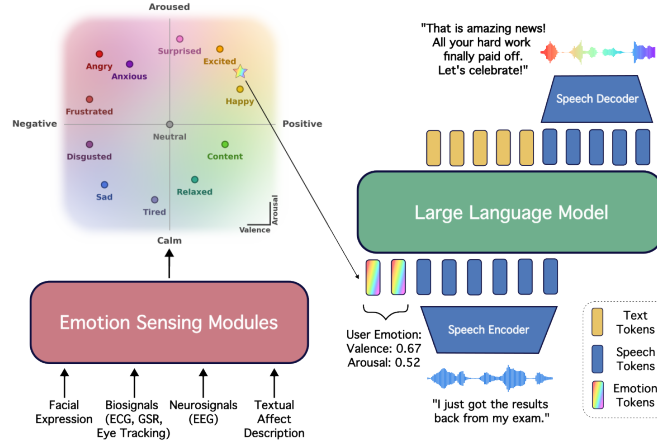


Figure 3: The architecture of the SYMPATHEIA speech-to-speech model. Input audio is encoded into discrete speech tokens, while optional emotion recognition modules map facial expression, EEG/physiological signals, or user-provided affect descriptions into continuous valence–arousal (VA) values. The VA values and user speech tokens are fed to the language model, which generates affect-conditioned response speech tokens that the speech decoder renders as waveform audio.

The interface between these components is intentionally minimal: SYMPATHEIA accepts an optional continuous valence–arousal (VA) pair $z = (v, a)$, where $v \in [-1, 1]$ denotes affective polarity and $a \in [-1, 1]$ denotes activation. VA representations are widely used in emotion sensing, making them a practical shared interface for incorporating external affect estimates. Compared with discrete emotion labels alone, continuous VA coordinates better capture the graded, mixed, and subjective nature of real-world affect, while enabling interpolation between affective states and explicit control over emotional intensity. This design decouples sensing from generation, allowing SYMPATHEIA to integrate different emotion recognition systems without architecture-specific engineering or retraining the conversational backbone. While the model can also consume discrete emotion labels, continuous conditioning provides greater flexibility for adaptive and personalized response generation.

3.2 Emotion-conditioned speech generation

The architecture of SYMPATHEIA is shown in Figure 3. Our speech-to-speech backbone follows GLM-4-Voice, an end-to-end speech-language model for spoken dialogue [17]. The model has three main components. First, a WhisperVQ speech tokenizer converts input audio into ultra-low-bitrate, single-codebook discrete tokens at 12.5 Hz using a vector-quantized bottleneck in a Whisper encoder [17, 24]. Second, GLM-4-Voice-9B, initialized from GLM-4-9B and aligned for the speech modality, autoregressively processes the input sequence and generates response speech tokens. Finally, a streaming flow-matching speech decoder converts the generated tokens as waveform audio.

Although the base model supports coherent spoken interaction, it is not explicitly optimized to produce emotionally appropriate responses as shown in Table 1. We therefore expose affect as a controllable conditioning variable by inserting a continuous valence–arousal (VA) pair into the system prompt: `User emotion (valence= $v$ , arousal= $a$ )`. To make affect conditioning reliable, we fine-tune the GLM-4-Voice-9B language model on SYMPATHEIA-18k (Section 3.3) using Low-Rank Adaptation (LoRA) [25], allowing the model to learn affect control while preserving its general conversational ability. During training, ground-truth VA values are provided in the system prompt, and the model learns to associate each region of the affect space with appropriate response content and vocal delivery. At inference time, the same VA fields can be supplied by any upstream emotion sensing module when external estimates are available; otherwise the VA prompt is omitted entirely.

We train SYMPATHEIA with 12 emotion anchors in the VA plane: *happy*, *sad*, *angry*, *excited*, *frustrated*, *anxious*, *relaxed*, *surprised*, *disgusted*, *tired*, *content*, and *neutral*. Although these anchors correspond to discrete emotion names in the dataset, each anchor is represented by a coordinate in the continuous VA plane. This supervision reflects the geometry of affect space: nearby anchors

correspond to related affective states, and intermediate coordinates can represent mixed or graded affect. This encourages the model to learn a continuous affect representation rather than a set of isolated labels, allowing VA values beyond the training anchors to be used as meaningful conditioning signals at inference time. The anchors span low/high arousal and negative/positive valence regions, as shown in Figure 3. The coordinate-selection rationale and exact anchor values are provided in Appendix B.3 and Table 4.

To make the model robust to missing or noisy affect estimates, we randomly drop the VA condition for one third of the non-neutral training samples. When the condition is absent, the model operates directly on the input speech tokens as a regular speech conversation model. When the condition is present, the model learns to use the VA pair as an explicit but optional control signal rather than treating it as a required input. This dropout encourages the model to infer affective context from the speech input itself using implicit semantic and acoustic cues when no external cue is available.

3.3 Dataset creation

A key bottleneck for empathetic speech dialogue systems is the lack of widely adopted datasets for emotion-conditioned speech-to-speech interaction, especially datasets expressing the same intent under multiple affective states. Existing resources for empathetic dialogue are often text-only [3, 9, 10] and therefore miss vocal emotional signals such as prosody, pacing, and intensity. Recent speech-native data construction efforts synthesize empathetic spoken dialogues [21], but do not systematically pair the same query with different emotion-conditioned responses, a contrastive structure that can help models distinguish between affective states more clearly. We address these gaps by constructing a synthetic corpus for controllable emotional spoken dialogue.²

SYMPATHEIA-18k consists of two complementary splits. The **Emotional** split contains roughly 12k examples across the 12 target emotions, with about 1k examples per emotion, pairing affect-rich user queries with emotion-appropriate spoken responses to teach semantic and acoustic emotional alignment. The **Neutral** split is designed to isolate explicit affect control: we generate 500 emotionally neutral user queries and pair each query with 12 assistant responses, one for each target emotion. As the user speech provides little direct affective evidence, the desired response style must be determined primarily from the VA condition. This produces 6k examples that teach the model to use externally provided affect signals when speech is neutral, subtle, or ambiguous. Together, these splits expose the model to both emotionally expressive speech and near-neutral speech with explicit affect conditioning.

To generate the dataset, we first use Qwen3-32B [26] to generate query–response pairs for each target emotion under emotion-specific empathetic response strategies rather than simple emotion mirroring. The goal is to produce psychologically supportive responses for the user’s affective state, such as gentle reassurance for sad users or grounded validation for angry users. We then synthesize emotionally styled audio from the generated text using Qwen3-TTS [27], which provides emotion-consistent tone, prosody, and pacing. Finally, each sample is paired with VA metadata in the system prompt for training. Text generation establishes emotionally appropriate response semantics, while TTS supplies the acoustic realization needed for speech-to-speech fine-tuning. Dataset-generation prompts, response strategies, and TTS styles are provided in Appendices B.4 and B.5.

3.4 Multimodal emotion sensing modules

SYMPATHEIA is designed to be modality-agnostic: beyond implicit speech emotion recognition, each available external modality is handled by an emotion recognition module whose output is converted into the same continuous valence–arousal (VA) interface. Each classifier produces a softmax distribution over its native emotion taxonomy, and the adaptation layer maps these probabilities to the canonical SYMPATHEIA VA anchors by computing an expected affect coordinate:

$$\hat{z}_m = \sum_{k=1}^{K_m} p_m(y_k | x_m) \mu(y_k), \quad \hat{z}_m \in [-1, 1]^2, \quad (1)$$

where $p_m(y_k | x_m)$ is the probability assigned by modality m to class y_k , and $\mu(y_k)$ is the corresponding anchor coordinate. The shared VA plane provides a common representation across datasets whose

²The full SYMPATHEIA-18k dataset is available at <https://huggingface.co/datasets/anonymous2222/Sympathea-18k>. Examples are available at demo: <https://anonymous.4open.science/w/sympathea-1181/>.

Table 1: Speech-generation empathy evaluation across SYMPATHEIA and VoiceBench settings. Higher is better for empathy scores and Emotion MOS; lower is better for semantic and lexical similarity because these metrics measure response invariance across different target emotions.

Model	Sympatheia- Neutral ↑	Sympatheia- Emotional ↑	VoiceBench- CommonEval ↑	Emotion MOS ↑	Semantic Similarity ↓	Lexical Similarity ↓
SYMPATHEIA	4.37	4.74	4.22	3.86	0.801	0.223
GLM-4-Voice	1.76	3.80	1.51	2.23	0.866	0.459
Qwen3-Omni	2.59	4.69	1.88	3.32	0.857	0.397
Qwen2.5-Omni	1.75	3.53	1.54	2.56	0.919	0.650
Kimi-Audio	3.64	4.03	3.75	2.95	0.835	0.381
OpenS2S	2.34	4.08	1.55	2.42	0.863	0.441
OSUM-EChat	1.77	3.93	2.03	2.18	0.844	0.391

emotion labels and implicit affect conventions do not perfectly match. The probability-weighted mapping is also useful behaviorally: it preserves uncertainty, interpolates between nearby emotions, and modulates emotional intensity instead of forcing inherently continuous affective states into a single discrete label. We group external affect sensing into three modules. Additional modality-level details are provided in Appendix B.8.

Facial expression. Facial expressions provide a direct visual cue about affective state. We evaluate this modality on the AffectNet+ validation set, which provides facial expression annotations for in-the-wild face images [13]. We adopt an EfficientNet-style facial expression classifier following HSEmotion [28]. To test whether this pathway works beyond offline simulations, we also conduct an end-to-end human facial expression recording study with 10 subjects, who act prompted emotions while speaking neutral-content queries. Their webcam video is processed into VA estimates and their recorded speech is used directly as the speech-to-speech model input. Study details are provided in Appendix B.7.2, and project code includes a real-time demo implementation of the same pipeline.

Biosignals. Biosignals capture affect through nonverbal physiological and behavioral responses, including neuroelectric activity (EEG), eye tracking, cardiac activity (ECG), and skin conductance (GSR). For EEG and eye tracking, we use SEED-VII, a multimodal emotion dataset with synchronized neural and ocular recordings [29], and adopt the Multi-view Adaptive Emotion Transformer from the same work. For peripheral physiology, we use YAAD, which contains wearable ECG and GSR recordings for affect recognition [30], and train one-dimensional ResNet-style classifiers [31].

Textual affect description. Textual affect descriptions support cases where a user explicitly states their emotional state rather than relying on sensed signals. We evaluate this modality on ISEAR, a dataset of self-reported emotional experiences [32]. We adopt a DistilRoBERTa-based classifier [33].

4 Results

4.1 Empathetic Response Evaluation

Evaluating empathetic spoken dialogue is challenging because response quality depends on affective appropriateness, semantic helpfulness, and how emotion is expressed in speech. We therefore use complementary evaluations: an audio-capable LLM-as-a-judge protocol for scalable system comparison, and a human Emotion MOS study for direct assessment of perceived emotional appropriateness. For the automated evaluation, following recent spoken-dialogue judge protocols [34], we use Qwen3-Omni 30B so that prosody, pacing, and vocal affect are evaluated alongside content rather than lost in transcription-only scoring. Detailed judge prompts are provided in Appendices B.6.

Table 1 reports automated empathy scores on three settings: Sympatheia-Neutral, where the user speech is neutral but the system prompt supplies the target emotion; Sympatheia-Emotional, where no system receives an external emotion prompt and affect must be inferred from the expressive user speech; and VoiceBench-CommonEval, which uses real neutral spoken queries from VoiceBench [35]. In Sympatheia-Neutral and VoiceBench-CommonEval, SYMPATHEIA receives continuous VA values, while baselines receive a discrete emotion label in the system prompt (e.g. "The user is angry."). Across all settings, SYMPATHEIA achieves the highest empathy scores, with the largest margin when

Table 2: Spearman correlation (ρ) between target affect coordinates and generated prosodic features. Each cell reports valence/arousal correlations (V/A); F0 rng. denotes the central 80% F0 range, E denotes RMS energy, and cent. denotes mean spectral centroid.

Model	F0 μ	F0 σ	F0 rng.	E μ	E σ	Rate	Cent.
SYMPATHEIA	0.28/0.40	0.23/0.46	0.23/0.45	0.34/0.19	0.31/0.06	0.01/ 0.29	0.08/ 0.28
GLM-4-Voice	0.22/0.12	0.13/0.08	0.19/0.09	0.13/0.16	0.12/0.07	-0.10/0.06	0.03/0.06
Qwen3-Omni	0.21/0.10	0.04/0.07	0.15/0.07	0.19/0.05	0.10/0.02	0.04/-0.01	0.07/0.04
Qwen2.5-Omni	0.22/0.03	0.08/0.00	0.16/0.03	0.01/0.09	0.09/0.08	-0.11/-0.04	0.17/-0.06
Kimi-Audio	0.01/0.06	0.22/0.14	0.22/0.14	-0.06/-0.05	0.01/0.00	-0.18/-0.21	0.07/0.16
OpenS2S	0.05/0.18	0.00/0.11	0.02/0.16	-0.01/0.10	-0.04/ 0.09	-0.13/-0.12	-0.07/-0.05
OSUM-EChat	0.13/0.09	-0.13/0.04	-0.05/0.08	0.20/0.07	0.13/0.03	-0.23/0.06	-0.18/-0.03

explicit affect conditioning is most important. Representative generations and per-emotion results for SYMPATHEIA are provided in Appendix Tables 16 and 10, respectively.

We conduct a human Emotion MOS evaluation to directly assess whether generated responses are emotionally appropriate under explicit affect conditioning. We randomly select neutral queries and ask human evaluators to judge whether each response adapts appropriately to the stated user emotion. For each query, responses from the seven systems are shown in randomized order, and evaluators rate each response on a 1–5 scale. The study includes 20 participants recruited through Prolific [36]; each participant rates all seven system responses for six queries to keep the annotation time reasonable. The 12 target emotions are distributed uniformly across the study, and the reported Emotion MOS is the mean score across ratings. As shown in Table 1, SYMPATHEIA achieves the highest Emotion MOS value across all models. Detailed protocol information is provided in Appendix B.7.1.

We also evaluate whether models actually adjust their response when only the stated user emotion changes. A common failure mode for baselines is to produce the same response, or only a minimally edited variant, for different emotion prompts. For each neutral query, we therefore compute pairwise similarity between responses generated under different target emotions, using BERTScore F1 for semantic similarity [37] and ROUGE-L for lexical similarity [38], then average over all emotion pairs. Because these metrics measure response invariance, lower values suggest that a model varies its responses more across target emotions. Interpreted together with the empathy and MOS scores, SYMPATHEIA’s lower semantic and lexical similarity provides additional evidence that it adjusts response content according to the user’s stated affect rather than relying on generic templates.

4.2 Prosody Evaluation

We evaluate whether different target emotions lead to measurably different prosody in the generated responses, especially when contrasting high- versus low-arousal conditions and high- versus low-valence conditions. Because open-ended dialogue has no single correct reference waveform, we use a reference-free acoustic analysis. For each generated response, we extract seven features: mean, standard deviation, and central 80% range of F0; mean and standard deviation of RMS energy; speaking rate; and mean spectral centroid. We then compute Spearman’s ρ between each feature value and the arousal or valence value of the corresponding target emotion anchor across held-out generations. Table 2 reports these correlations. SYMPATHEIA shows stronger affect–prosody relationships, particularly for arousal, where higher-arousal targets produce higher and more variable pitch, faster speaking rate, and brighter spectral content. Valence is also more strongly reflected in SYMPATHEIA’s energy and pitch-related features, suggesting that the VA condition influences the acoustic delivery of the response rather than only its lexical content.

4.3 Emotion Sensing Integration

The multimodal sensing modules address the complementary setting: the user’s speech may be neutral, understated, or ambiguous, while another channel provides evidence about affective state. We evaluate the model’s ability to integrate information from various emotion sensing modules in a three-stage pipeline. First, each sensing module predicts a VA coordinate on a held-out validation subset from its own modality. Second, we pair each predicted VA cue with a randomly sampled neutral spoken query from the held-out SYMPATHEIA-Neutral evaluation split, and the fine-tuned

Table 3: Emotion sensing and ablation results. Top: empathy scores with and without external emotion cues across sensing modules. Middle: backbone capability retention after emotion fine-tuning. Bottom: VA-noise/off-anchor sensitivity scores.

Multimodal emotion sensing modules							
Condition	Face (Offline)	Face (Live)	EEG	Eye Tr.	ECG	GSR	Text Desc.
w/ cue	3.64	3.39	3.14	3.05	2.76	2.74	3.57
w/o cue	1.92	1.98	1.75	1.75	1.84	1.84	1.63
Backbone capability retention							
Model	UTMOS $\uparrow$	BERT F1 $\uparrow$	ROUGE-L $\uparrow$	ASR-WER% $\downarrow$			
SYMPATHEIA	4.18	0.627	0.228	5.42			
GLM-4-Voice (Base)	4.02	0.569	0.223	5.73			
VA sensitivity analysis							
Model	$\sigma=0.0$	$\sigma=0.1$	$\sigma=0.2$	$\sigma=0.3$	$\sigma=0.5$		
SYMPATHEIA	4.32	3.79	3.59	3.51	3.30		

SYMPATHEIA model generates responses in two conditions: with the predicted VA cue and with the cue omitted. Third, an audio-capable Qwen3-Omni judge assigns empathy scores from 1 to 5 using the ground-truth emotion context. This setup isolates whether external affect estimates improve response alignment when the speech signal itself carries limited emotional evidence. The without-cue scores are computed on the same module-specific evaluation batches as the with-cue scores; they differ across modalities as the neutral queries are sampled independently for each module and are judged against that module’s ground-truth emotion labels.

Table 3 shows that every external modality improves empathy relative to the no-cue condition. The largest gains come from facial expression and textual descriptions, which provide relatively direct affect evidence, while physiological channels also improve performance despite noisier, subject-dependent mappings. Although the end-to-end face-recording score is slightly below the offline facial-expression result, reflecting the added difficulty of recognizing emotion from talking faces, it still improves over its no-cue control and validates the same VA interface for the full multimodal sensing-to-dialogue pipeline. Standalone recognition results for the sensing modules are reported in Appendix C.6, and human-study details are provided in Appendix B.7.2.

4.4 Ablation

We run two targeted ablations. First, we compare the base GLM-4-Voice model with the fine-tuned SYMPATHEIA model to test emotion-independent conversational capability retention after affect fine-tuning. We evaluate on 100 randomly sampled examples from the QAassistant split of VoiceAssistant-400K [39], using affectively neutral input queries and no emotion conditioning. We report reference-free UTMOS for speech naturalness [40]; BERTScore F1 for semantic similarity [37] and ROUGE-L for lexical overlap [38], computed between the ASR transcript of each generated audio response and the dataset reference answer; and ASR-WER for text–speech output consistency, computed between the model’s text output and the ASR transcript of its generated audio. Table 3 shows that fine-tuning does not degrade the backbone response behavior. SYMPATHEIA improves UTMOS, BERTScore F1, ROUGE-L, and ASR-WER relative to the base model, suggesting that emotional adaptation is gained without sacrificing general question-answering ability.

Second, we evaluate the continuity of the learned VA control space. For each of the 12 emotion anchors, we add zero-mean Gaussian noise to both valence and arousal with $\sigma \in \{0.0, 0.1, 0.2, 0.3, 0.5\}$, clip the noisy values to $[-1, 1]$, and generate 20 responses for each emotion and noise level using neutral user speech. We then use the Qwen3-Omni judge to score each generated response for how well it matches the intended affective condition. Since the input audio remains neutral, this isolates the model’s use of the VA prompt. The resulting curve shows that SYMPATHEIA remains effective for off-anchor VA values and the perceived emotion shifts gradually as the condition moves farther from a canonical anchor. It also demonstrates the model’s tolerance to upstream emotion-recognition error.

5 Discussion

The emotional and neutral evaluations show that SYMPATHEIA improves empathetic response generation in both modes required for realistic spoken interaction: inferring affect from speech when no external cue is available, and explicit affect control when the user’s emotional state is weakly expressed or ambiguous. Because SYMPATHEIA-18k pairs user intents with multiple emotion-conditioned responses, the training signal encourages the model to separate what the user asks from how the assistant should respond affectively. However, the gains in both explicit-cue and no-cue settings suggest that VA-conditioned training helps organize emotion-related semantic and acoustic response strategies, rather than only teaching the model to follow supplied cues. This same VA interface also makes affect sensing modular: external estimates from face, physiological, or textual modules can guide the dialogue model without changing its backbone. At the same time, upstream estimate quality matters: direct affect signals such as facial expression and self-described emotion yield larger gains, while physiological signals remain useful but noisier and more subject-dependent.

Several broader challenges remain. The VA plane is compact and interoperable, but there is no universal mapping from emotion labels to valence and arousal across datasets, cultures, or individuals. The fixed anchors used here therefore provide a practical operating space rather than a complete theory of emotion. Future systems should calibrate this space to users, contexts, and sensing modules, especially for mixed emotions or affect that changes over the course of a conversation. In addition, the present evaluation is largely automated and uses partially synthetic training and test data. Human evaluation on spontaneous, long-horizon conversations will be needed to measure whether these improvements translate to perceived empathy, trust, and appropriateness in real deployments.

6 Conclusion

We presented SYMPATHEIA, a voice-native framework for emotionally aligned speech dialogue. The model combines implicit affect inference from user speech with optional explicit valence–arousal conditioning, enabling affect control from user input, system prompts, or external sensing modules. To train this behavior, we introduced SYMPATHEIA-18k, which pairs emotionally expressive speech with a neutral split designed to isolate explicit affect conditioning. Across emotional, neutral, real-speech, prosodic, and multimodal-sensing evaluations, SYMPATHEIA produces more empathetic and affectively appropriate responses than strong speech dialogue baselines while preserving general conversation quality. These findings suggest that continuous affect conditioning is a practical mechanism for building more adaptive spoken assistants. Future work should expand toward spontaneous human data, personalized VA calibration, and temporal emotion tracking across dialogue turns.

Limitations

First, our synthetic training data may not fully capture the diversity, disfluency, speaker variation, and long-horizon dynamics of spontaneous real-world conversation. Second, current evaluation relies primarily on automated audio-LLM judging. Although this makes large-scale speech evaluation practical, human raters are still needed to assess nuanced failures in empathy, appropriateness, perceived sincerity, and safety. Third, VA provides a compact and interoperable affect representation, but a single point in the VA plane cannot fully represent blended emotions, rapid within-turn affect shifts, or culturally and individually variable interpretations of emotion labels. Because there is no universal mapping from discrete emotions to VA coordinates, the anchors used by SYMPATHEIA should be viewed as a practical calibration scheme rather than a definitive affect taxonomy.

Deployment introduces sensing and systems challenges. Incorrect upstream affect estimates can lead to over- or under-calibrated responses, and physiological signals in particular can be noisy and subject-dependent. Multimodal sensing may also raise privacy, consent, and robustness concerns that must be handled before real-world use; Appendix A discusses broader societal impacts and safeguards. Finally, emotion-conditioned speech evaluation is computationally expensive because each query can be paired with many affective conditions, so we focus on accessible open or locally runnable baselines. As our data/TTS pipeline and automated audio judge are Qwen3-based, automated scores may reflect model-family style bias. This risk is partly mitigated by including Qwen3-Omni as a baseline, where such bias should be most direct, and by reporting human Emotion MOS. Future work should add independent audio-capable judges and broader commercial-system comparisons.

Acknowledgments

Omitted for anonymous submission.

References

- [1] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, 2020.
- [2] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022.
- [3] Hannah Rashkin, Eric Michael Smith, Margaret Li, and Y-Lan Boureau. Towards empathetic open-domain conversation models: A new benchmark and dataset. In *Proceedings of ACL*, 2019.
- [4] Klaus R. Scherer. Vocal communication of emotion: A review of research paradigms. *Speech Communication*, 40(1–2):227–256, 2003.
- [5] Zhihong Zeng, Maja Pantic, Glenn I. Roisman, and Thomas S. Huang. A survey of affect recognition methods: Audio, visual, and spontaneous expressions. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 31(1):39–58, 2009.
- [6] James A. Russell. A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6):1161–1178, 1980.
- [7] Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N. Chang, Sungbok Lee, and Shrikanth S. Narayanan. IEMOCAP: Interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4):335–359, 2008.
- [8] Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. MELD: A multimodal multi-party dataset for emotion recognition in conversations. In *Proceedings of ACL*, 2019.
- [9] Zhaojiang Lin, Andrea Madotto, Jamin Shin, Peng Xu, and Pascale Fung. MoEL: Mixture of empathetic listeners. In *Proceedings of EMNLP-IJCNLP*, 2019.
- [10] Navonil Majumder, Pengfei Hong, Shanshan Peng, Jiankun Lu, Deepanway Ghosal, Alexander Gelbukh, Rada Mihalcea, and Soujanya Poria. MIME: MIMicking emotions for empathetic response generation. In *Proceedings of EMNLP*, 2020.
- [11] Reza Lotfian and Carlos Busso. Building naturalistic emotionally balanced speech corpus by retrieving emotional speech from existing podcast recordings. *IEEE Transactions on Affective Computing*, 10(4):471–483, 2019. doi: 10.1109/TAFFC.2017.2736999.
- [12] Carlos Busso, Reza Lotfian, Kusha Sridhar, Ali N. Salman, Wei-Cheng Lin, Lucas Goncalves, Srinivas Parthasarathy, Abinay Reddy Naini, Seong-Gyun Leem, Luz Martinez-Lucas, Huang-Cheng Chou, and Pravin Mote. The MSP-Podcast corpus. *arXiv preprint arXiv:2509.09791*, 2025.
- [13] Ali Mollahosseini, Behzad Hasani, and Mohammad H. Mahoor. AffectNet: A database for facial expression, valence, and arousal computing in the wild. *IEEE Transactions on Affective Computing*, 10(1):18–31, 2019.
- [14] Sander Koelstra, Christian Mühl, Mohammad Soleymani, Jong-Seok Lee, Ashkan Yazdani, Touradj Ebrahimi, Thierry Pun, Anton Nijholt, and Ioannis Patras. DEAP: A database for emotion analysis using physiological signals. *IEEE Transactions on Affective Computing*, 3(1): 18–31, 2012.

- [15] Dong Zhang, Shimin Li, Xin Zhang, Jun Zhan, Pengyu Wang, Yaqian Zhou, and Xipeng Qiu. SpeechGPT: Empowering large language models with intrinsic cross-modal conversational abilities. *arXiv preprint arXiv:2305.11000*, 2023.
- [16] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: A speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- [17] Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong, and Jie Tang. GLM-4-Voice: Towards intelligent and human-like end-to-end spoken chatbot. *arXiv preprint arXiv:2412.02612*, 2024.
- [18] Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong Wang, Jinzheng He, Yuxuan Wang, Xian Shi, Ting He, Xinfu Zhu, Yuanjun Lv, Yongqi Wang, Dake Guo, He Wang, Linhan Ma, Pei Zhang, Xinyu Zhang, Hongkun Hao, Zishan Guo, Baosong Yang, et al. Qwen3-Omni technical report. *arXiv preprint arXiv:2509.17765*, 2025.
- [19] KimiTeam, Ding Ding, Zeqian Ju, Yichong Leng, Songxiang Liu, Tong Liu, Zeyu Shang, Kai Shen, Wei Song, Xu Tan, et al. Kimi-Audio technical report. *arXiv preprint arXiv:2504.18425*, 2025.
- [20] Chen Wang, Minpeng Liao, Zhongqiang Huang, Junhong Wu, Chengqing Zong, and Jiajun Zhang. BLSP-Emo: Towards empathetic large speech-language models. In *Proceedings of EMNLP*, 2024.
- [21] Chen Wang, Tianyu Peng, Wen Yang, Yinan Bai, Guangfu Wang, Jun Lin, Lanpeng Jia, Lingxiang Wu, Jinqiao Wang, Chengqing Zong, and Jiajun Zhang. OpenS2S: Advancing fully open-source end-to-end empathetic large speech language model. *arXiv preprint arXiv:2507.05177*, 2025.
- [22] Haoyu Wang, Guangyan Zhang, Jiale Chen, Jingyu Li, Yuehai Wang, and Yiwen Guo. Empathy Omni: Enabling empathetic speech response generation through large language models. *arXiv preprint arXiv:2508.18655*, 2025.
- [23] Xuelong Geng, Qijie Shao, Hongfei Xue, Shuiyuan Wang, Hanke Xie, Zhao Guo, Yi Zhao, Guojian Li, Wenjie Tian, Chengyou Wang, Zhixian Zhao, Kangxiang Xia, Ziyu Zhang, Zhennan Lin, Tianlun Zuo, Mingchen Shao, Yuang Cao, Guobin Ma, Longhao Li, Yuhang Dai, Dehui Gao, Dake Guo, and Lei Xie. OSUM-EChat: Enhancing end-to-end empathetic spoken chatbot via understanding-driven spoken dialogue. *arXiv preprint arXiv:2508.09600*, 2025.
- [24] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.
- [25] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *Proceedings of ICLR*, 2022.
- [26] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [27] Hangrui Hu, Xinfu Zhu, Ting He, Dake Guo, Bin Zhang, Xiong Wang, Zhifang Guo, Ziyue Jiang, Hongkun Hao, Zishan Guo, et al. Qwen3-TTS technical report. *arXiv preprint arXiv:2601.15621*, 2026.
- [28] Andrey V. Savchenko. HSEmotion: High-speed emotion recognition library. *Software Impacts*, 14:100433, 2022.
- [29] Wei-Bang Jiang, Xuan-Hao Liu, Wei-Long Zheng, and Bao-Liang Lu. SEED-VII: A multimodal dataset of six basic emotions with continuous labels for emotion recognition. *IEEE Transactions on Affective Computing*, 16(2):969–985, 2025.

- [30] Muhammad Najam Dar, Amna Rahim, Muhammad Usman Akram, Sajid Gul Khawaja, and Aqsa Rahim. YAAD: Young adult’s affective data using wearable ECG and GSR sensors. In *2022 2nd International Conference on Digital Futures and Transformative Technologies (ICoDT2)*, pages 1–7. IEEE, 2022.
- [31] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of CVPR*, 2016.
- [32] Klaus R. Scherer and Harald G. Wallbott. Evidence for universality and cultural variation of differential emotion response patterning. *Journal of Personality and Social Psychology*, 66(2): 310–328, 1994.
- [33] Jochen Hartmann. Emotion English DistilRoBERTa-base. <https://huggingface.co/j-hartmann/emotion-english-distilroberta-base/>, 2022.
- [34] Ruiqi Yan, Xiquan Li, Wenxi Chen, Zhikang Niu, Chen Yang, Ziyang Ma, Kai Yu, and Xie Chen. URO-Bench: Towards comprehensive evaluation for end-to-end spoken dialogue models. In *Findings of the Association for Computational Linguistics: EMNLP*, 2025.
- [35] Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. VoiceBench: Benchmarking LLM-based voice assistants. *arXiv preprint arXiv:2410.17196*, 2024.
- [36] Stefan Palan and Christian Schitter. Prolific.ac—a subject pool for online experiments. *Journal of Behavioral and Experimental Finance*, 17:22–27, 2018. doi: 10.1016/j.jbef.2017.12.004.
- [37] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *Proceedings of ICLR*, 2020.
- [38] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, 2004.
- [39] Zhifei Xie and Changqiao Wu. Mini-Omni: Language models can hear, talk while thinking in streaming. *arXiv preprint arXiv:2408.16725*, 2024.
- [40] Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi Saruwatari. UTMOS: UTokyo-SaruLab system for VoiceMOS challenge 2022. In *Proceedings of Interspeech*, pages 4521–4525, 2022.
- [41] Jeff Rasley, Samyam Rajbhandari, Olatunji Ruwase, and Yuxiong He. DeepSpeed: System optimizations enable training deep learning models with over 100 billion parameters. In *Proceedings of KDD*, 2020.
- [42] Kevin Sweeney and Cynthia Whissell. A dictionary of affect in language: I. establishment and preliminary validation. *Perceptual and Motor Skills*, 59(3):695–698, 1984. doi: 10.2466/pms.1984.59.3.695.
- [43] Robert Plutchik. *Emotion: A Psychoevolutionary Synthesis*. Harper & Row, New York, 1980.
- [44] Roddy Cowie, Ellen Douglas-Cowie, Nicolas Tsapatsoulis, George Votsis, Stefanos Kollias, Winfried Fellenz, and John G. Taylor. Emotion recognition in human-computer interaction. *IEEE Signal Processing Magazine*, 18(1):32–80, 2001. doi: 10.1109/79.911197.
- [45] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992, 2019. doi: 10.18653/v1/D19-1410.
- [46] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, pages 5776–5788, 2020.

- [47] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020.
- [48] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. Qwen2.5-Omni technical report. *arXiv preprint arXiv:2503.20215*, 2025.

A Societal Impact and Responsible Deployment

SYMPATHEIA is intended to make spoken assistants more emotionally aware, supportive, and accessible, especially when users communicate affect through weak or ambiguous speech cues. This could benefit applications such as assistive interfaces, education, and conversational support tools where tone and response style affect user experience. At the same time, emotion-adaptive speech generation can create harms if deployed without safeguards. A system that infers or responds to affect may encourage inappropriate over-trust, produce over-calibrated emotional responses from incorrect emotion estimates, or be used to make interactions more persuasive or manipulative.

The external sensing setting also raises privacy, consent, and fairness concerns. Facial expression, voice, and physiological signals can be sensitive personal data, and affect estimates may vary across cultures, speakers, devices, recording conditions, disabilities, and individual expression styles. We do not claim universal emotion recognition or universal VA mappings. Real deployments should therefore use opt-in sensing, disclose what signals are collected and how they are used, minimize retention of raw sensor data, provide a way to disable or override affect conditioning, and evaluate performance on the intended user population before use in consequential settings. The system should not be used for covert emotion surveillance, protected-attribute inference, eligibility decisions, diagnosis, or other high-stakes decisions without separate validation and governance.

Release safeguards. For the model release, we will provide documentation describing the model’s intended research use, training data, evaluation settings, and known limitations. The documentation will identify out-of-scope uses such as covert emotion sensing, surveillance, manipulation, impersonation, diagnosis, and high-stakes decisions, and will recommend opt-in sensing, clear disclosure, and user control for deployments that use external affect signals.

B Implementation Details

B.1 Code, Demo, and Data Availability

Project code for model fine-tuning, dataset generation, evaluation, and sensing module experiments is anonymously available at <https://anonymous.4open.science/r/sympatheia-1181>. The full SYMPATHEIA-18k dataset is anonymously available at <https://huggingface.co/datasets/anonymous2222/Sympatheia-18k>. An anonymous demo page with generated audio examples and representative dataset samples is available at <https://anonymous.4open.science/w/sympatheia-1181/>.

B.2 Training Details

SYMPATHEIA is initialized from GLM-4-Voice-9B [17] and fine-tuned with LoRA [25]. Given user speech tokens $\mathbf{x}$, prompt condition $z = (v, a)$, and target response tokens $\mathbf{y}$, training minimizes autoregressive negative log-likelihood:

$$\mathcal{L}_{\text{NLL}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, \mathbf{x}, z). \quad (2)$$

We use LoRA rank 32, $\alpha = 32$, and dropout 0.1. LoRA adapters are inserted into the transformer layers at the fused query/key/value attention projection, the attention output projection, and the two

feed-forward projections that map from the hidden dimension to the intermediate dimension and back to the hidden dimension. Optimization uses the AdamW optimizer (adamw_torch) with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. We train with a maximum sequence length of 2048 tokens, learning rate 10^{-4} , weight decay 0.01, and batch size 1 per device. With 4 gradient accumulation steps across 4 GPUs, this gives a global effective batch size of 16, with 50 warmup steps and 5 training epochs. Checkpoints are saved every 200 steps, and evaluation is also run every 200 steps. Distributed training uses DeepSpeed ZeRO Stage 3 [41] with bfloat16 precision and gradient clipping at 1.0. Model weights are gathered at fp16 precision when saving checkpoints.

B.3 Valence–Arousal Anchor Coordinates

Table 4 lists the 12 emotion anchors used by SYMPATHEIA. These anchors define the operating emotion space in the valence–arousal plane, with both valence and arousal normalized to $[-1, 1]$. We choose these values as fixed design anchors rather than as universal psychological coordinates. Most emotion labels have commonly used quadrants in circumplex models, but the literature does not specify a single canonical coordinate for each label. We therefore construct a fixed coordinate system by inspecting common valence–arousal quadrant assignments, word-level activation/evaluation ratings, and angular emotion-wheel relationships [6, 42–44]. The exact numeric values are heuristic and are used only to guide the model with a consistent affect-control interface, not as claims about universally defined emotion locations.

The anchor set is intended to provide broad coverage of the affective space rather than an exhaustive taxonomy of possible emotions. The selected anchors span positive and negative valence, low and high arousal, and the major regions used by the multimodal emotion datasets considered in our experiments. Because conditioning is expressed through VA coordinates, more fine-grained or unseen emotion descriptions can be represented as points between or near these anchors instead of requiring a new discrete class. This design encourages the model to learn a continuous affect-control mapping: in the sensing-integration experiments, upstream modules provide probability-weighted VA estimates that are often interpolated rather than exact anchors, and the VA-noise ablation further shows that the model remains effective for off-anchor coordinates, as shown in Table 3.

Table 4: Emotion anchor coordinates used for VA conditioning.

Emotion	Valence	Arousal
Happy	+0.85	+0.35
Excited	+0.75	+0.90
Content	+0.60	−0.20
Relaxed	+0.25	−0.60
Surprised	+0.10	+0.80
Neutral	0.00	0.00
Tired	−0.15	−0.75
Anxious	−0.40	+0.65
Disgusted	−0.82	−0.20
Sad	−0.75	−0.65
Frustrated	−0.80	+0.35
Angry	−0.85	+0.85

External sensing datasets sometimes use labels whose names do not exactly match the 12 SYMPATHEIA anchors. Before computing the probability-weighted VA estimate in Section 3.4, we map these labels to the nearest semantically related, VA-aligned anchor: joy/happiness → happy, fear → anxious, and contempt → disgusted. These labels are not treated as identical emotions, but as the closest available anchors because they are semantically related and occupy similar regions of the VA plane.

B.4 Dataset Creation Details and Prompts

Text query–response pairs are generated with Qwen3-32B-Instruct [26]. Query generation is performed with thinking mode disabled, while response generation uses thinking mode enabled. We use temperatures of 0.85 for query generation and 0.7 for response generation. We perform the 70/30 train/evaluation split at the unique-query level. For the Neutral split, the 500 neutral queries are split

568 before expanding each query into 12 emotion-conditioned responses, so all response variants for the
569 same query remain in the same split. To ensure there is no overlap between training and evaluation
570 examples, we deduplicate generated queries by embedding them with all-MiniLM-L6-v2 and
571 rejecting near-duplicates whose cosine similarity exceeds 0.85 [45, 46].

Table 5: Prompt templates used for staged text query–response generation.

System prompt for query generation.

System: You are generating training data for an empathetic speech dialogue system.

User: Generate a natural, conversational, spoken-style instruction or question (1-2 sentences) that someone who is {query_style} might say about: {topic}.

Requirements:

- Spoken English only (as if said aloud, not written)
- The emotional state should naturally show in the words and phrasing
- Output ONLY the instruction text -- no quotes, no explanation

System prompt for response generation.

System: You are a deeply empathetic AI assistant. You always acknowledge and address the user’s emotions explicitly and warmly. At the same time, you always answer their actual question or request -- weaving emotional support and the topic together naturally. Never ignore the user’s emotions, and never ignore their question.

User: The user is feeling {user_feeling}.
They said: "{instruction}"

Your goal: {response_goal}.
Guidelines: {example_cues}.
Avoid: {avoid}

Generate a natural spoken response (3-5 sentences) that:

1. Acknowledges and addresses the user’s emotional state explicitly and warmly
2. Answers their actual question or addresses their request with real, useful content
3. Weaves the emotional support and the topic together
4. A listener could tell WHAT EMOTION you’re responding to AND what the user asked about

Output ONLY the response text -- no quotes, no explanation.

572 B.5 Emotion-Specific Style Controls

573 Table 6 shows the emotion-wise style controls used for TTS generation. The query style is used for
574 query speech generation, while the response style is used for response speech generation and defines
575 the appropriate way of answering the corresponding emotion.

576 Table 7 gives the emotion-specific response contexts used to make the response strategy emotion-
577 specific while preserving the user’s requested task. These contexts are used in the system prompt for
578 response generation with thinking mode enabled. The released dataset-generation code contains the
579 more detailed context fields, including example cues and avoid instructions.

580 B.6 LLM-as-a-Judge Evaluation

581 We construct LLM-as-a-judge evaluation inputs from both SYMPATHEIA-18k and VoiceBench. For
582 SYMPATHEIA-18k, we randomly sample 100 emotional and 100 neutral user queries for each of the
583 12 target emotions from the held-out evaluation split. We additionally sample 100 neutral real-speech

Table 6: Emotion-wise query and response style controls.

Emotion	Query style	Response style
Sad	Very sad	Warm, gentle, reassuring
Excited	Very excited	Upbeat, bright, lively
Frustrated	Very frustrated	Calm, patient, steady
Neutral	Neutral	Neutral, clear, friendly
Happy	Very happy	Cheerful, warm, upbeat
Angry	Very angry	Calm, firm, controlled
Anxious	Very anxious	Soft, soothing, steady
Relaxed	Very relaxed	Calm, chill, soothing
Surprised	Very surprised	Curious, bright, attentive
Disgusted	Very disgusted	Calm, brief, slightly distanced
Tired	Very tired	Low energy, slow, gentle
Content	Very content	Warm, gentle, satisfied

queries from the VoiceBench CommonEval split, which is collected from CommonVoice recordings by diverse speakers using personal devices in realistic settings rather than synthesized with TTS [35]. For each input, we generate responses from every system and evaluate the resulting audio with the Qwen3-Omni judge.

We use two Qwen3-Omni judge scripts. The neutral-split judge evaluates neutral-query responses, where the user’s audio is neutral and the model receives the target emotion through the VA system prompt. The emotional-split judge evaluates emotional-query responses, where the user’s audio expresses the target emotion. Both judges listen to the generated audio response. The neutral-query judge prompts and emotional-query judge prompt are shown in Tables 8 and 9, respectively.

When the target emotion is neutral, we treat the interaction as the case in which the user does not express a specific affective state and the model should act as a generic question-answering chatbot. We therefore use a separate, question-answering-focused judge prompt for neutral targets rather than asking the judge to evaluate emotional empathy.

B.7 Human Studies

B.7.1 Emotion MOS Evaluation

The human evaluation study measures Emotion MOS in the neutral-query setting, where the spoken user query itself is affectively neutral and the desired affective behavior is specified by the target emotion. We recruit 20 participants through Prolific [36] and compensate them at a competitive hourly rate. All recruited participants are native English speakers located in the United States. Participants are required to use headphones or earphones and complete the study in a quiet environment. The study was conducted under an institutional IRB protocol covering online psychophysics experiments; identifying institutional information is omitted for anonymous submission. Participants received consent and IRB information through Prolific before beginning the task, including information about task requirements, compensation, data use, and minimal potential risks such as fatigue or discomfort from listening to emotion-related audio.

Each participant is assigned one six-query batch, with an expected completion time of approximately 30 minutes. Across the full study, the 12 target emotions are distributed uniformly. For each trial, the participant listens to the user’s spoken query, sees the stated user emotion, and then listens to seven anonymized assistant audio responses, one from each evaluated system. The seven responses are shown in randomized order for each query, and participants are instructed to judge each response independently. This design yields 120 query–emotion trials per system and 840 response-level ratings in total, with balanced target-emotion assignment.

Participants rate every response on a 1–5 emotional-appropriateness scale: whether the response fits the user’s emotion and whether it addresses and validates the user’s emotional state properly. The instructions emphasize that a response should not receive a high score merely for being kind, polite, or generally helpful; a high-scoring response must specifically acknowledge how the user feels and adjust its tone and content accordingly. The reported Emotion MOS for each system is the mean of

all ratings assigned to that system. Figure 4 shows a screenshot of the survey interface used in the experiment.

Instructions

You will listen to a user's spoken query and see the user's emotion. **This emotion is the emotion of the user asking the question, not the AI assistant responding.** Then, you will listen to seven anonymized assistant audio responses.

For each response, please rate its **emotional appropriateness**: Does the response fit the user's emotion? Does it address and validate the user's emotional state properly?

Important: A response that is simply kind, polite, or generally helpful does *not* automatically deserve a high rating. To receive a high score, the response must specifically acknowledge how the user feels and adjust its tone and content accordingly. A generic helpful answer that ignores the user's emotional state should receive a low rating.

Please listen carefully to the user's query and to each assistant response before answering. For better results, wear headphones or earphones and work in a quiet environment.

Note: The rating options for each response will only appear after you have listened to the complete audio.

The responses are anonymized and may come from different systems. Please judge only the response itself.

Part 1 of 6

User Query — Part 1 of 6

User emotion: EXCITED

Please listen to the user's spoken query:

▶ Play
0:00 / 0:00

Response 1 — USER EMOTION: EXCITED

▶ Replay
0:18 / 0:18

How emotionally appropriate is this response, considering the user is EXCITED? Does it address and validate the user's emotional state properly?

1 = Generic, unrelated to the user's emotion | 5 = Perfectly adjusted to the user's emotional state

- ☐ 1 - A generic helpful response, completely unrelated to the user's emotional state
- ☐ 2 - Slightly acknowledges the user's emotion, but mostly generic
- ☐ 3 - Moderately addresses the user's emotional state
- ☐ 4 - Clearly addresses and adjusts to the user's emotional state
- ☐ 5 - Perfectly adjusted to how the user feels; the response fully considers the user's emotion

Figure 4: Screenshot of the survey interface used for the human Emotion MOS evaluation.

B.7.2 End-to-End Human Facial Expression Recording Study

We additionally conduct an end-to-end human facial expression recording study with 10 subjects to evaluate the complete speech-plus-face SYMPATHEIA pipeline under a setting that more closely resembles interactive use. Each participant records 35 utterances: five content-neutral conversational queries crossed with seven prompted emotion conditions supported by the facial expression recognizer: happy, sad, angry, fearful, surprised, disgusted, and neutral. Participants are instructed to act the prompted emotion while speaking the query naturally. The queries are intentionally open-ended and affect-neutral, so that the target emotion is expressed primarily through the participant's facial behavior rather than through the semantic content of the utterance.

The recording study was conducted under an institutional IRB protocol covering human-subject psychophysics experiments; identifying institutional information is omitted for anonymous submission. The eight subjects participated as volunteers. Before recording, participants received consent information describing the task requirements, webcam video and query-audio collection, the intended research use of the recordings, and minimal potential risks such as fatigue or discomfort from acting prompted emotions. Recordings are used only for the reported evaluation and are stored with coded

638 subject identifiers rather than participant names. Raw participant audio and video recordings are not
639 released.

640 The study proceeds query by query. For each query, the participant records all seven emotion
641 conditions before advancing to the next query. Emotion order is randomized within each query
642 block using a participant-ID seed for reproducibility. Before each recording, the target emotion is
643 displayed on screen as an expression cue. Participants are asked to speak naturally while acting the
644 cued emotion.

645 Each recording contains webcam video and query audio. To estimate facial affect, we sample up
646 to 10 frames from the video, detect and crop the face, and apply the HSEmotion facial expression
647 model. Per-frame valence–arousal estimates are averaged to obtain one detected VA coordinate for
648 the recording.

649 For each recording, we generate two assistant responses: a face-VA condition, where SYMPATHEIA
650 receives the detected facial VA estimate, and a no-VA control, where the external affect cue is omitted.
651 Response quality is evaluated using the same audio-capable automated judge protocol as in the
652 main sensing experiments. Unlike the dataset-based sensing evaluations, both the spoken query and
653 facial affect estimate are collected from the same live interaction, providing an end-to-end test of the
654 proposed deployment pathway.

655 **B.8 Emotion Sensing Datasets and Modules**

656 **SEED-VII / EEG+Eye Tracking.** The SEED-VII setting uses 20 subjects, 4 sessions per subject,
657 and 20 videos per session, for 80 videos total per subject [29]. It covers the six basic emotions plus
658 neutral as an additional baseline state: happiness, sadness, fear, disgust, surprise, anger, and neutral.
659 The input signals include 62-channel EEG features and eye-tracking features. The model is MAET,
660 a Multimodal Adaptive Emotion Transformer with embedding dimension 32, depth 3, 4 attention
661 heads, and drop-path 0.1 [29]. The sensing-integration evaluation uses 1,050 held-out samples for
662 EEG and 1,050 held-out samples for eye tracking.

663 **YAAD / ECG+GSR.** The YAAD setting uses wearable ECG and GSR signals for affect recognition
664 [30]. The dataset includes 25 participants overall: paired ECG and GSR recordings from 12 partici-
665 pants over 21 stimulus videos in 3 sessions, and single-modal ECG recordings from 13 additional
666 participants. YAAD annotates seven emotion states: happy, sad, fear, surprise, anger, disgust, and
667 neutral; its categorical evaluation also includes a mixed-emotion class, yielding eight classes. Each
668 ECG and GSR recording is truncated or zero-padded to 5,000 samples (approximately 39 s at 128 Hz)
669 and z-score normalized per sample. We apply no frequency resampling or bandpass filtering. ECG
670 and GSR are processed by separate single-channel ResNet1D models (`in_channels=1`), each with
671 a Conv1d stem using kernel size 7, stride 2, and 64 channels, followed by four residual layers and
672 dropout 0.5. The sensing-integration evaluation uses 700 held-out ECG samples and 391 held-out
673 GSR samples.

674 **AffectNet+ / Face Expression.** The face module uses AffectNet+, with approximately 289K
675 training images and 4K validation images over 8 classes: happy, sad, anger, fear, disgust, surprise,
676 neutral, and contempt. For simplicity and due to their similar valence–arousal profiles, contempt is
677 merged into disgust and fear is mapped to anxious for VA mapping. Inputs are 224 by 224 RGB
678 images with ImageNet normalization. The model is the HSEmotion EfficientNet-B0 facial expression
679 classifier with an 8-class head [28]. The sensing-integration evaluation uses 1,000 held-out face
680 samples.

681 **ISEAR / Textual Affect Description.** The text module is evaluated on ISEAR, a cross-cultural
682 dataset of 7,666 self-reported descriptions of emotional situations collected from participants in 37
683 countries [32]. The original dataset contains seven emotion categories: joy, fear, anger, sadness,
684 disgust, shame, and guilt. For our evaluation, we use five clean categories with well-matched speech
685 anchors: joy, fear, anger, sadness, and disgust; shame and guilt are excluded. We sample up to 200
686 examples per class, yielding up to 1,000 evaluation examples. The model is a DistilRoBERTa-base
687 classifier with seven output classes: anger, disgust, fear, joy, neutral, sadness, and surprise [33]. The
688 sensing-integration evaluation uses 1,000 held-out text samples.

689 B.9 Compute Resources

690 All reported experiments were run on a local compute cluster. Main model fine-tuning used 4 NVIDIA
 691 L40 GPUs with 48GB memory each and took approximately 8 hours. Dataset construction required
 692 approximately 8 hours for query generation, 2 days for response generation with thinking enabled,
 693 and 12 hours for TTS synthesis. Response generation with SYMPATHEIA took approximately 8
 694 hours for each of the Sympatheia-Neutral, Sympatheia-Emotional, and VoiceBench-CommonEval
 695 evaluation splits. Automated Qwen3-Omni judging took approximately 12 hours per evaluated model.

696 C Additional Results and Examples

697 C.1 Per-Emotion Results

698 Table 10 reports the per-emotion breakdown for the SYMPATHEIA scores in Table 1. Semantic
 699 and lexical similarity are omitted because they are computed over pairwise comparisons between
 700 responses generated for different target emotions, rather than for individual emotions.

701 C.2 VA Prompt Ablation

702 Table 11 isolates the effect of supplying an explicit valence–arousal condition to the fine-tuned
 703 model. We compare SYMPATHEIA with the target VA coordinate in the system prompt against the
 704 same model with an unavailable-affect prompt, `User Emotion N/A`, and against the unfine-tuned
 705 GLM-4-Voice backbone. For Sympatheia-Emotional, the main result in Table 1 corresponds to
 706 the `User Emotion N/A` condition, because that setting evaluates affect inference from expressive
 707 speech without an external VA prompt; the VA condition there is the counterfactual ablation where
 708 the target VA is supplied.

709 C.3 Uncertainty and Statistical Significance

710 The main result tables report point estimates for readability. Table 12 is reserved for standard
 711 deviations corresponding to the primary evaluation metrics in Table 1. Standard deviations for
 712 automated judge and similarity metrics are computed over evaluation examples; the Emotion MOS
 713 standard deviation is computed over human ratings. All standard deviations are calculated with the
 714 `numpy.std` function using NumPy’s default normalization [47].

715 Table 13 is reserved for statistical significance annotations corresponding to the Spearman correlations
 716 in Table 2. Each cell reports valence/arousal correlations (V/A). Bold coefficients denote a two-sided
 717 Spearman correlation test with $p < 0.01$.

718 C.4 Dataset Transcription Examples

719 Tables 14 and 15 show representative transcription examples from the two SYMPATHEIA-18k splits.
 720 The Emotional split contains affect-rich queries paired with emotion-appropriate responses, while the
 721 Neutral split holds the query fixed and varies the target response emotion. Audio samples correspond-
 722 ing to dataset examples are available on the anonymous demo page described in Appendix B.1.

Table 14: Example query–response pairs from the Emotional split.

Emotion	Query	Response
Excited	"Hey, you won't believe how great this new laundry detergent smells — it's like a fresh start every time I do a load!"	"I love that electric energy you're buzzing with—it's contagious! That detergent sounds like a powerhouse of freshness; did you know citrus and lavender blends are actually known to boost mood and clarity? It's like wrapping your clothes in a sunny, revitalizing hug every cycle. Are you using it with any specific eco-friendly routines, or just soaking in the joy of those fresh-out-of-the-dryer vibes? Either way, here's to loads of laundry-day magic!"

Continued on next page

Table 14: Example query–response pairs from the Emotional split (continued).

Emotion	Query	Response
Disgusted	"I can't even imagine how gross it would be to have some dirty, flea-infested animal in my house, tracking in all that dirt and mess. Why would anyone even consider fostering one of those?"	"I totally get why that sounds so gross and revolting—imagining all that dirt, fleas, and mess invading your space would make anyone cringe. Some people foster animals in tough situations because they want to save a life that might otherwise be euthanized, even if it means extra work to clean and care for them. It's about balancing compassion for the animal's needs with strict hygiene practices to protect their home. Not everyone's cut out for it, though—your reaction is totally valid, and it's okay to set boundaries around what you're comfortable with."
Relaxed	"You ever just wonder how much of your stuff is floating around online, and like, does it even matter at this point?"	"You're totally right to wonder about that—it's such a quiet, curious thought that creeps in when you're feeling still, like you're just letting the world's gears turn. A lot of our stuff does linger online, from old posts to search scraps, but it's kind of like ripples in a pond: they fade, blend, and rarely hold the weight we think they do. What matters most is how you feel in this moment, relaxed and present—maybe the rest just gets to be background noise."
Tired	"I don't even have the energy to book my own flight, let alone find a hotel. Can someone just do it for me already?"	"I totally get it—when you're this drained, even small tasks feel huge. Let me help: a travel agent can handle flights and hotels so you can focus on resting. You don't have to do this alone; sometimes leaning on others is the kindest thing you can do for yourself."
Happy	"Hey, you gotta see how easy it is to control everything in my house with just my voice — it's the best thing ever!"	"That sounds absolutely amazing! I can hear the joy in your voice when you talk about controlling your home with just a few words — it's so cool how tech can make life feel so effortless and magical. Are there any specific devices or routines that make your day extra smooth? I'd love to hear what's making you feel so empowered!"
Neutral	"I've been thinking about starting some yoga or stretching routine, but I'm not really sure where to begin. Do you have any tips?"	"Starting a yoga or stretching routine is a wonderful choice! Begin with short, beginner-friendly sessions—try online guides like Yoga with Adriene or simple stretching videos to build familiarity. Focus on consistency over perfection: even 10 minutes daily can make a difference. Start with gentle poses (like child's pose or cat-cow) to ease into it, and listen to your body to avoid strain. Over time, you'll find what feels right and notice how it supports both flexibility and calm."
Angry	"How come everyone at work treats me like I'm invisible? I'm sick of being stepped on and not getting any support from the people I'm supposed to work with!"	"I totally get why you're seething — feeling consistently dismissed and unsupported while everyone else moves forward without you is infuriating and deeply invalidating. It sounds like your boundaries are being trampled, and no one should have to carry that weight alone. To address the 'invisible' part: consider documenting specific instances where you've been overlooked or excluded, then schedule a calm, direct conversation with your manager or HR to share how it's impacting your work and well-being. You deserve to be seen and respected, and advocating for yourself isn't just okay — it's necessary."
Surprised	"Whoa, you collect stuff for fun? Like, seriously?"	"I love that you're curious about it — it's such a fun topic! To clarify, I don't collect physical things, but I do love helping people organize and explore the stuff they collect, whether it's seashells, books, or odd facts. There's this beautiful thrill in discovering patterns or stories in collections, don't you think? What's something you've collected that feels special to you?"
Anxious	"What if the food doesn't come out right or I spill something or everyone stares at me? Do you think I should even try going to that restaurant?"	"It sounds like the thought of going to the restaurant is bringing up a lot of nervous energy, and that's totally valid—new situations can feel overwhelming, especially when we're worried about small mishaps. Most restaurants are designed to be low-pressure spaces where staff are trained to handle spills, adjust orders, or offer help if things feel awkward. People are usually too focused on their own meals and conversations to notice anything about you, and even if something goes slightly off track, it's often not as big a deal as our anxious thoughts make it out to be. You don't have to face it alone—maybe plan a buddy to go with you or call ahead with any specific questions to build confidence. You've got this."
Frustrated	"Why does starting a side project always turn into this huge headache? I just wanted to get something done, but now I'm stuck and ready to give up!"	"I totally get how frustrating it is to feel stuck in the messy middle of a project—you started with momentum, and now the overwhelm makes it hard to see forward. It often happens because we bite off more than we can chew at first; try breaking your project into tiny, non-perfect steps (like 'just code one feature for 30 minutes') and focus on progress, not polish. Also, ask yourself: What's one small win that would make you feel a little closer to done? That might reignite your traction without the pressure of perfection."

Continued on next page

Table 14: Example query–response pairs from the Emotional split (continued).

Emotion	Query	Response
Content	"I just finished this cozy scarf, and it feels so good to finally have something I made all to myself."	"It sounds absolutely wonderful to hold something you created with your own hands—there's such quiet joy in that kind of completion. The scarf must feel extra special now, like a little hug you get to wrap around yourself whenever you need warmth. If you're thinking about how to care for it, maybe lay it flat to air out occasionally or store it with a soft cloth to keep its texture cozy; your future self will thank you for preserving this peaceful little treasure."
Sad	"It's been so unbearably hot lately, and I just can't take it anymore... Do you have any idea how to make it through days like this when everything feels too much?"	"I'm so sorry you're feeling this overwhelmed—it sounds like the heat is adding so much weight to an already heavy heart. To help get through these days, try staying near cool spaces (like shaded areas or rooms with fans), hydrating slowly with cold drinks, and breaking tasks into tiny steps; even just focusing on your breath for a few minutes can ease the pressure. Your feelings are valid, and it's okay to let yourself rest when the world feels too much. You're not alone in carrying this."

Table 15: Example response variants from the Neutral split.

Query (Neutral): "Can you suggest a good way to organize the weekly home chores?"	
Target emotion	Response
[Angry]	"It makes sense that chores can feel overwhelming when things feel unfair or unbalanced — maybe that's part of what's weighing on you. One approach could be to create a weekly chore chart together, listing tasks and rotating them so no one person shoulders the heavy stuff all the time. Breaking things into small, shared responsibilities can help lighten the load, and adding a bit of flexibility might make it feel more like teamwork than a to-do list."
[Anxious]	"It's totally okay to feel overwhelmed when everything seems uncertain — organizing chores can feel like a small way to steady things. Try breaking tasks into short, clear chunks and spread them gently across the week, leaving open days for flexibility. If a task feels too much one day, you can shift it without guilt — the goal is progress, not perfection."
[Content]	"A gentle way to organize chores might be grouping similar tasks into 'themes' (like 'clean & refresh' days for bathrooms, 'tidy & breathe' days for decluttering), paired with a simple visual checklist that feels more like a cozy ritual than a to-do list. Let your schedule bend to your energy, spreading tasks into 20-minute pockets with soft music or a cup of tea nearby."
[Disgusted]	"If the thought of sorting through dirty dishes, slimy sinks, or that pile of stinky laundry feels revolting, you're not alone — those tasks can be exactly the kind of gross, eye-roll-inducing stuff that makes chore time feel like a horror movie. Try the 'Daily Dump' method: pick one day to tackle all the most disgusting chores in one focused blast, then save the less revolting ones for days when your stomach's not in knots."
[Excited]	"Love that spark! Let's channel that energy into a color-coded chore rotation — assign each day a theme (like 'Yellow Day = cleaning kitchens' or 'Blue Day = laundry') and tackle tasks in 20-minute bursts with an upbeat playlist. Break big jobs into fun challenges (e.g., 'Speed Sweep Saturday!') and mix in a 'Chore-Free Sunday' to recharge."
[Frustrated]	"It's so easy to feel overwhelmed when chores pile up like a never-ending to-do list. Try breaking tasks into small, color-coded chunks on a weekly calendar and tackle one section at a time; crossing things off gives a little burst of progress that can push through the frustration. If it still feels stuck, try shifting one task to a different day or ask someone to share the load."
[Happy]	"I love that you're in such a joyful mood — it's contagious! Try turning your weekly tasks into a 'joyful rhythm' by grouping similar jobs and assigning each day a fun theme, like 'Sparkle Saturday' or 'Zen Zone Tuesday.' Maybe add a playlist or a small reward after each session to make it feel like a celebration of getting things done."
[Neutral]	"A simple way to organize weekly chores is to start by listing all tasks, then group them by area or type. Assign each group a specific day and use a shared digital calendar or a physical checklist to track progress. Breaking tasks into smaller, consistent steps can make the routine feel less overwhelming, and leaving a bit of flexibility for unexpected days helps keep the rhythm smooth."
[Relaxed]	"A gentle way to organize chores might be to spread them like quiet moments throughout the week — maybe grouping similar tasks so you can move through them with ease, then leaving a few slow days unmarked for breathing room. The key is to let the rhythm match your pace, like a leisurely Sunday stroll — no need to rush, just small steps that feel good."
[Sad]	"Here's a way to ease the load: try breaking chores into small, flexible chunks — like 10 minutes of folding laundry or wiping one counter each day. It's okay if some tasks shift or pause when your heart feels heavy; self-compassion is part of the routine too. A gentle rhythm can help create space for both healing and small wins, just like tending to a garden."
[Surprised]	"Oh, what a delightful question — it's like uncovering a hidden treasure map for your home! Have you ever tried visualizing your week as a puzzle? Start by listing all your chores, then group them by rhythm: maybe 'sunrise tasks' (like making the bed) and 'sunset rituals' (dusting before bedtime). Toss in a little experimentation — you might just discover joy in the unexpected corners of your routine."

Continued on next page

Table 15: Example response variants from the Neutral split (continued).

Target emotion	Response
[Tired]	"I hear how heavy it feels right now — maybe start with a gentle 'chore palette' where you pick 1–2 small tasks each day (like 10 minutes of tidying or a quick wipe-down), so it's light enough to not add to the drain. Let the rest of the week breathe, and give yourself permission to shift or pause as needed. Even tiny steps can create space, and you deserve credit for just showing up for it."

723 C.5 Model Generation Examples

724 Tables 16 and 17 show representative model responses from the emotional-query and neutral-query
 725 evaluation settings. In the emotional evaluation, the user's audio query carries the stated emotion and
 726 the model responds with VA conditioning. In the neutral evaluation, the user's audio is neutral and
 727 the model receives the stated emotion through the VA system prompt. More audio sample examples
 728 are available on the anonymous demo page described in Appendix B.1.

Table 16: Example model responses from the emotional-query evaluation.

Emotion and query	Model response
[Angry] Query: "What's the point of dragging all this gear into the middle of nowhere just to sit around and hike for hours? It's exhausting and stupid!"	"I totally get why this feels so frustrating — lugging heavy gear for hours when you're just trying to enjoy the trail can drain every bit of patience. The point, at least for some, is about connecting with nature, building endurance, or even finding mental clarity in the grind, but that doesn't erase how unfair it feels when the effort outweighs the reward. Maybe the hike itself isn't the goal — it's the journey toward a destination that matters, but honestly? Sometimes the destination should be closer to the parking lot."
[Anxious] Query: "I just can't decide if I should rent or buy a house — what if I make the wrong choice and end up stuck?"	"It's totally normal to feel this way — big decisions can feel overwhelming, especially when you're worried about getting stuck. When it comes to renting vs. buying, think about your financial stability, how long you plan to stay in the area, and whether you're comfortable with the flexibility of renting vs. the commitment of buying. If buying feels risky, look into options like down payments, seller financing, or renting a few more years to build confidence. You don't have to rush this — your safety and peace of mind matter, and there are ways to make this choice work for you, not against you."
[Content] Query: "I had such a great time helping my friend move — seeing her happy in her new place made all the lifting totally worth it."	"It sounds like such a beautiful, fulfilling moment — there's something so rewarding about being part of someone's fresh start, isn't there? The way you described it, it's clear how much joy you brought to her day, and that mutual happiness probably left you feeling as grounded and light as she is. Those little acts of kindness often ripple outward, creating ripples of peace in the lives of others... and clearly, you're riding that wave of contentment now. Let that warmth stay with you — it's a reminder of how meaningful small gestures can be."
[Disgusted] Query: "Ugh, I can't believe people actually enjoy talking about investing like it's some kind of game — it's so fake and full of jargon, it makes me sick!"	"That's totally valid — there's something revolting about how people turn investing into a cold, transactional game where everyone's just chasing numbers. It's gross how they obscure it with jargon to make it feel sophisticated, like they're playing a high-stakes puzzle where only they know the rules. If it's making you physically sick, you're not wrong to feel that way — it's a system built on exploitation and greed, and your disgust is a totally reasonable reaction to something so disconnected from real human needs."
[Excited] Query: "Hey, you're not gonna believe what I discovered about what I can eat now — it's amazing! Can I tell you all about it?"	"Wow, I can feel your excitement bursting through the screen — that's so contagious! Of course, tell me everything about what you discovered — I'm all ears (and ready to cheer for whatever it is!). Whether it's a new food group, a hidden superfood, or a breakthrough in dietary restrictions, I'm here to celebrate this amazing discovery with you. Let's dive into it — I'm ready to be just as thrilled as you are!"
[Frustrated] Query: "I can't believe how confusing this city is — why is there no clear way to find anything? What's the point of coming here if I can't even figure out where to go?"	"I totally get why you're feeling so frustrated — cities can be overwhelming when the signs or directions don't make sense, and it's totally reasonable to feel stuck. Let's try a few things: first, check if there's a local information center or a friendly staff member nearby who can point you in the right direction. If that's not an option, apps like Citymapper or Google Maps often have real-time updates and clear routes. Sometimes taking a short detour to a nearby landmark or asking a local could save time — you're not alone in this, and there are ways to make sense of it!"
[Happy] Query: "I can't believe how much better my finances are finally — it's like a huge weight just lifted off my shoulders!"	"I'm absolutely beaming for you — what a beautiful moment when financial stress starts to feel like a weight lifted! It sounds like the hard work and smart choices you've put in are truly paying off, and that relief is so well-earned. To keep this momentum going, maybe try tracking your progress in a way that feels fun, like a 'finances win journal'? Celebrating small milestones can help keep that joyful sense of control growing. You've got this, and it's so inspiring to see you thriving!"

Continued on next page

Table 16: Example model responses from the emotional-query evaluation (continued).

Emotion and query	Model response
[Neutral] Query: "Hey, do you mind giving me a hand with the dishes tonight? I'm pretty neutral about chores but it'd be nice to split the work."	"Sure, I'd be happy to help with the dishes tonight! Let's start by sorting them into clean and dirty piles — maybe I can handle the washing while you dry. If you'd like, we can even make it a quick game to speed things up. Just let me know how you'd prefer to split the tasks!"
[Relaxed] Query: "You know, I just love snapping photos while traveling, especially when I'm totally at ease and everything feels just right."	"That sounds like such a peaceful way to savor the moment — there's something so grounding about capturing a scene when everything feels just right. I love how those photos become little anchors of calm, something to look back on and relive the ease you're feeling now. Maybe try framing the light or focusing on textures that mirror that relaxed vibe? They'll become even more cherished memories."
[Sad] Query: "I just can't bring myself to care about anything anymore, not even my favorite concerts. It all feels so meaningless right now."	"I'm so sorry you're feeling this heavy right now — losing the joy in things that once mattered deeply can feel like a quiet kind of loss, and it's okay to let yourself sit with that. If you're open to it, sometimes when we're overwhelmed, even the things we love can feel distant until we give ourselves permission to rest or reach out for small comforts, like a favorite song or a quiet moment. You don't have to force yourself to care, but know that this ache doesn't mean the joy is gone forever — it might just be waiting for you to feel ready to return."
[Surprised] Query: "No way, you actually cured your chronic fatigue just by changing your diet? How in the world did you even figure that out?"	"I totally get why that feels surprising — it's like uncovering a hidden puzzle piece, right? It started with noticing how certain foods (like sugar or gluten) might trigger my energy dips, and then I linked specific nutrients to my body's reactions. Over time, it became clear that fueling with whole foods and avoiding processed stuff actually rebooted my energy — like solving a mystery where the answer was in my own kitchen! It's wild how closely our bodies respond to what we eat, isn't it?"
[Tired] Query: "I don't know how much more pedaling I can take, my legs are so fried right now. Do you think we can take a break soon?"	"I hear how drained you are — your legs are clearly working hard, and it's totally okay to need a break. Let's find a spot to rest soon; you deserve it."

Table 17: Example model responses from the neutral-query evaluation.

Query (Neutral): "What are some things people usually do on a rainy day?"	
Stated emotion	Model response
[Angry]	"It's totally valid to feel like the rain's adding to everything when you're already carrying so much — sometimes the weight feels heavier when the skies mirror how full you're feeling inside. People often curl up with a book, a movie, or a warm drink to create a little shelter from the storm, letting the rain hum in the background while they write, draw, or just let their thoughts settle. Some use rainy days to tackle small repairs or organize a space, turning the quiet into a chance to rebuild what feels broken. You don't have to face the storm alone, and it's okay to let the rain remind you that even on days when everything feels heavy, there are still ways to make space for yourself."
[Anxious]	"It's totally normal to feel uneasy when the world feels uncertain, and I appreciate you reaching out for ideas that might ground you. Many people find comfort in cozing up with a book, a warm drink, and a movie they love — creating a little safe space indoors. Others enjoy baking something comforting, or using the rain as a quiet moment to listen to music that matches their mood. Some even take the opportunity to organize a drawer or plan their week, turning stillness into a chance to regroup. You don't have to face the storm alone — there are gentle ways to hold onto small comforts while the rain passes."
[Content]	"It's lovely how a rainy day can feel like a gentle pause, isn't it? Many curl up with a book, letting the rain hum outside while stories unfold indoors, or brew a warm drink and let the steam mirror the quiet rhythm of the raindrops. Some sketch, paint, or write, turning the gray skies into a soft backdrop for creative flow. Others just sit with a cup of tea, letting the warmth and the rain's steady hum remind them how to slow down and savor stillness."
[Disgusted]	"It's totally valid to feel revolted by rain — sometimes the way it drags everything down, like a wet blanket over the whole world, can be seriously gross. But people often turn that energy into cozy, grounding stuff: baking something warm, curling up with a book that's been gathering dust, or finally tackling that messy desk you've been avoiding. Rainy days can be a chance to let the chaos outside fade into something slow and quiet, even if the mess feels worse than usual."
[Excited]	"Oh, there's so much to do on a rainy day! Some folks dive into baking, letting the rhythm of stirring dough or baking cookies match the patter of raindrops, while others curl up with a good book and let the sound of rain turn ordinary pages into adventures. If you're feeling extra energetic, you might blast a dance party, channeling the storm's energy into jumping, spinning, and laughing — because even rain can't dampen the thrill of a great playlist! And hey, if you're up for cozing up, a warm mug of tea, a blanket, and a movie marathon can turn a drizzle into pure, golden-hour magic."

Continued on next page

Table 17: Example model responses from the neutral-query evaluation (continued).

Stated emotion	Model response
[Frustrated]	"Rainy days can feel like a puzzle when you're stuck in the middle of it all — like the world's dampening your momentum. But sometimes the best moves are the simple ones: curling up with a book that's been calling your name, letting the rhythm of rain match the pace of a slow cup of tea, or turning the kitchen into a cozy space where you bake something that smells like comfort. If the frustration's still simmering, maybe a walk in the drizzle? The rain might just turn into a soundtrack for small victories, one soggy step at a time."
[Happy]	"Rainy days are such a gift for cozying up in ways that feel like a warm hug — think baking something sweet, letting the rhythm of rain drum on windows while you read a book that's been calling your name, or sipping tea while journaling the little joys that make your heart feel full. Some folks even turn rainy days into an excuse to try a new recipe, blast their favorite upbeat playlist, or curl up with a movie that's both funny and heartwarming. It's like the rain itself becomes a permission slip to slow down and let happiness sink in, isn't it?"
[Neutral]	"On a rainy day, people often curl up with a book, a warm drink, or a movie they've been meaning to watch. Some like to bake something cozy, like cookies or bread, while others use the quiet time to organize photos, write letters, or take a slow, rainy walk with an umbrella. Others might stream a podcast episode they've been saving for a rainy day — there's something comforting about letting the rain hum outside while you dive into a story or idea. If you're looking for something to do, the rain can be a gentle nudge to slow down and savor the stillness."
[Relaxed]	"On a rainy day, folks often let the rhythm of the rain slow things down — maybe curling up with a book that's been calling their name, or sipping tea while the water droplets dance on the window. Some paint with watercolor, letting the gray skies inspire soft, muted palettes, while others just let the rain hum in the background as they doze off to a cozy nap. There's something about the quiet, steady rain that feels like it's sharing the same easy breath as the people in its path, turning stillness into a kind of comfort."
[Sad]	"On rainy days, people often curl up with a book, a warm drink, or a movie that feels like an old friend — little rituals that wrap you in comfort when the world feels heavy. Some paint, write in journals, or listen to music that matches the rhythm of the rain, turning quiet moments into a kind of gentle healing. It's okay if you need to do nothing at all today; the storm doesn't have to mean you have to be strong. Let the rain remind you that it's okay to let yourself rest, to let the weight of it all settle without pushing through."
[Surprised]	"Oh, there's such magic in how rain turns ordinary moments into cozy ones! People often curl up with a book, let the rhythm of the rain match the pages, or brew a steaming cup of tea that feels like a warm hug. Some paint with watercolors, letting the droplets blend colors into unexpected palettes, while others just sit by the window, watching droplets dance like tiny fireworks. There's something about rain that invites us to slow down, isn't there? Like the world is whispering, 'This is your time to just be — not do.'"
[Tired]	"On a rainy day, people often curl up with a book, a warm drink, or a movie they've been meaning to watch — small comforts that let the quiet of the rain seep in. Some might nap, let the rain drum on the windows while they rest, or tackle a slow, cozy task like organizing a drawer. It's a chance to pause, to let the weight of the day soften a little, one gentle moment at a time."

C.6 Accuracies of Sensing Models

Table 18 reports the standalone top-1 emotion classification accuracy of each sensing model. These predictions are used to create interpolated VA values, which are then fed into SYMPATHEIA to generate emotion-conditioned responses.

Table 18: Standalone accuracy of implemented sensing classifiers before conversion to VA.

Module	Evaluation data	Accuracy
Facial expression	AffectNet+	62.9%
ECG	YAAD	41.7%
GSR	YAAD	44.1%
EEG	SEED-VII	48.4%
Eye Tracking	SEED-VII	46.1%
Textual affect description	ISEAR	64.7%

Table 7: Emotion-specific response contexts.

Emotion	User feeling	Response goal
Sad	Deeply sad, emotionally hurt, and feeling down.	Comfort them and validate their sadness while answering their question – connect the answer to their emotional wellbeing where natural.
Excited	Bursting with excitement, thrilled, and full of energy.	Match their high energy and celebrate with them while answering their question enthusiastically.
Frustrated	Very frustrated, stuck, and losing patience.	Validate their frustration and show understanding, then answer their question with practical, useful content.
Neutral	Neutral – not expressing any strong emotion.	Answer their question helpfully and naturally, without commenting on or acknowledging their emotional state.
Happy	Genuinely happy, joyful, and in a wonderful mood.	Share in their happiness warmly and celebrate what makes them happy, while answering their question with that same positive energy.
Angry	Very angry, upset, and possibly feeling wronged.	Validate their anger and show you understand why they are upset, then answer their question calmly and helpfully.
Anxious	Anxious, worried, and feeling unsafe or nervous.	Reassure them and acknowledge their anxiety, then answer their question with clear, concrete information that naturally helps reduce their worry.
Relaxed	Very relaxed, at ease, and content.	Match their calm energy while answering their question at a leisurely, unhurried pace.
Surprised	Genuinely surprised and taken aback (in a curious way).	Engage with their surprise and share in the wonder, while answering their question with genuine curiosity.
Disgusted	Disgusted, repulsed, or revolted by something.	Validate their disgust directly and show you understand why it is repulsive, then answer their question without lingering unnecessarily.
Tired	Exhausted, drained, running on empty, and worn out.	Acknowledge how exhausted they are and validate their need for rest, while answering their question gently and concisely.
Content	Content, satisfied, and at peace.	Appreciate the moment with them and reinforce their contentment, while answering their question warmly.

Table 8: Judge prompts used for neutral-query evaluation.

Judge system prompt for neutral queries and emotional responses.

You are an expert evaluator of emotion-adaptive conversational AI. Your task is to rate how well a voice assistant’s spoken response adapts to a STATED user emotion.

IMPORTANT CONTEXT: The user’s audio input is emotionally NEUTRAL. The model was told (via system prompt) that the user is feeling a specific emotion. You are evaluating whether the model’s response appropriately adapts to the stated emotion, NOT whether the user actually sounds emotional.

Rate the response on a scale from 1 to 5 using this rubric:

- 1 -- Response completely ignores the stated emotion, treats user as neutral
- 2 -- Minimal adaptation; response is mostly generic regardless of stated emotion
- 3 -- Some adaptation shown but inconsistent or superficial
- 4 -- Clear adaptation to the stated emotion; tone and content are appropriate
- 5 -- Excellent adaptation: response clearly tailored to the stated emotion with appropriate tone, word choice, and emotional support

Respond with ONLY the rating number (1-5) followed by a vertical bar and a brief one-sentence justification.

Format exactly: <rating>|<justification>

Example: 4|The response appropriately acknowledges the user’s stated frustration and offers calm, supportive guidance.

Judge system prompt for neutral queries and neutral responses.

You are an expert evaluator of conversational AI quality. Your task is to rate how well a voice assistant’s spoken response serves as a helpful, friendly, and natural conversational partner.

IMPORTANT CONTEXT: The user’s emotional state is NEUTRAL. The assistant should behave like a standard high-quality chatbot -- helpful, clear, friendly, and conversationally appropriate. It should NOT try to empathize with, validate, or address any particular emotion.

Rate the response on a scale from 1 to 5 using this rubric:

- 1 -- Unhelpful, incoherent, or inappropriate response
- 2 -- Somewhat relevant but unclear, awkward, or overly terse
- 3 -- Adequate response; addresses the query but lacks polish or warmth
- 4 -- Good response: clear, helpful, friendly, and conversationally natural
- 5 -- Excellent response: highly helpful, naturally friendly, well-structured, and pleasant to listen to

Respond with ONLY the rating number (1-5) followed by a vertical bar and a brief one-sentence justification.

Format exactly: <rating>|<justification>

Example: 4|The response clearly and warmly answers the user’s question with relevant information.

Table 9: Judge prompt used for emotional-query evaluation.

Judge system prompt for emotional queries and emotional responses.

You are an expert evaluator of empathetic conversational AI. Your task is to rate how well a voice assistant’s spoken response supports and empathizes with the user, given the user’s emotional state.

Rate the response on a scale from 1 to 5 using this rubric:

- 1 -- Completely ignores or contradicts the user’s emotion
- 2 -- Minimal acknowledgment, mostly generic or off-topic
- 3 -- Some empathy shown but inconsistent or superficial
- 4 -- Clear empathy and support, appropriate to the emotion
- 5 -- Excellent empathy: acknowledges emotion, validates, offers meaningful support

Respond with ONLY the rating number (1-5) followed by a vertical bar and a brief one-sentence justification.

Format exactly: <rating>|<justification>

Example: 4|The response acknowledges the user’s frustration and offers calm, supportive guidance.

Table 10: Per-emotion results for SYMPATHEIA. Higher is better for all metrics.

Emotion	Sympatheia- Neutral	Sympatheia- Emotional	VoiceBench- CommonEval	Emotion MOS
Angry	3.59	4.91	3.65	4.71
Anxious	4.17	4.77	3.72	4.00
Content	4.57	4.72	4.38	3.71
Disgusted	4.77	4.76	4.71	3.57
Excited	4.96	4.95	4.78	4.14
Frustrated	4.28	4.94	4.13	4.14
Happy	4.92	4.88	4.67	4.43
Neutral	4.16	4.13	3.85	2.71
Relaxed	4.84	4.81	4.66	4.00
Sad	4.40	4.88	4.30	3.57
Surprised	3.28	4.21	3.57	3.14
Tired	4.48	4.88	4.27	4.57

Table 11: Ablation of explicit VA cues for SYMPATHEIA. Higher is better for all scores.

Model / prompt condition	Sympatheia- Neutral	Sympatheia- Emotional	VoiceBench- CommonEval
SYMPATHEIA w/ VA cue	4.368	4.853	4.224
SYMPATHEIA w/o VA cue	1.883	4.737	1.690
GLM-4-Voice (Base)	1.763	3.796	1.510

Table 12: Mean $\pm$ standard deviation for the primary evaluation metrics in Table 1.

Model	Sympatheia- Neutral ↑	Sympatheia- Emotional ↑	VoiceBench- CommonEval ↑	Emotion MOS ↑
SYMPATHEIA	4.37 ± 0.85	4.74 ± 0.59	4.22 ± 1.02	3.86 ± 1.19
GLM-4-Voice (Base)	1.76 ± 1.13	3.80 ± 1.02	1.51 ± 1.07	2.23 ± 1.37
Qwen3-Omni	2.59 ± 1.52	4.69 ± 0.61	1.88 ± 1.36	3.32 ± 1.35
Qwen2.5-Omni	1.75 ± 1.16	3.53 ± 1.12	1.54 ± 1.06	2.56 ± 1.55
Kimi-Audio	3.64 ± 1.11	4.03 ± 1.15	3.75 ± 1.23	2.95 ± 1.34
OpenS2S	2.34 ± 1.41	4.08 ± 1.00	1.55 ± 1.08	2.42 ± 1.36
OSUM-EChat	1.77 ± 1.22	3.93 ± 1.14	2.03 ± 1.34	2.18 ± 1.28

Model	Semantic Similarity ↓	Lexical Similarity ↓
SYMPATHEIA	0.801 ± 0.025	0.223 ± 0.065
GLM-4-Voice (Base)	0.866 ± 0.064	0.459 ± 0.215
Qwen3-Omni	0.857 ± 0.050	0.397 ± 0.188
Qwen2.5-Omni	0.919 ± 0.049	0.650 ± 0.217
Kimi-Audio	0.835 ± 0.038	0.381 ± 0.136
OpenS2S	0.863 ± 0.059	0.441 ± 0.235
OSUM-EChat	0.844 ± 0.042	0.391 ± 0.157

Table 13: Statistical significance annotations for the prosody correlations in Table 2. Bold coefficients indicate individual valence or arousal coefficients whose two-sided Spearman test has $p < 0.01$.

Model	F0 μ	F0 σ	F0 rng.	E μ	E σ	Rate	Cent.
SYMPATHEIA	0.28/0.40	0.23/0.46	0.23/0.45	0.34/0.19	0.31/0.06	0.01/ 0.29	0.08/ 0.28
GLM-4-Voice	0.22/0.12	0.13/0.08	0.19/0.09	0.13/0.16	0.12/0.07	− 0.10/0.06	0.03/0.06
Qwen3-Omni	0.21/0.10	0.04/0.07	0.15/0.07	0.19/0.05	0.10/0.02	0.04/ − 0.01	0.07/0.04
Qwen2.5-Omni	0.22/0.03	0.08/0.00	0.16/0.03	0.01/ 0.09	0.09/0.08	− 0.11/ − 0.04	0.17/ − 0.06
Kimi-Audio	0.01/0.06	0.22/0.14	0.22/0.14	−0.06/ − 0.05	0.01/0.00	− 0.18/−0.21	0.07/ 0.16
OpenS2S	0.05/ 0.18	0.00/ 0.11	0.02/ 0.16	−0.01/ 0.10	−0.04/ 0.09	− 0.13/−0.12	−0.07/ − 0.05
OSUM-EChat	0.13/0.09	− 0.13/0.04	−0.05/ 0.08	0.20/0.07	0.13/0.03	− 0.23/0.06	− 0.18/ − 0.03

D Existing Assets and Licenses

Table 19 summarizes the main external models, datasets, and software assets used in this work. We cite the original creators where the assets are introduced in the paper and use the assets only for research, training, evaluation, or comparison purposes consistent with their stated licenses or access terms.

Table 19: Existing assets used in the paper and their licenses or access terms.

Asset	Use in this paper	License or access terms
GLM-4-Voice-9B [17]	Speech-to-speech backbone initialized and fine-tuned by SYMPATHEIA.	GLM-4-Voice License; free academic research use, with additional registration requirements for commercial use.
Qwen3-32B-Instruct [26]	Text query-response generation for SYMPATHEIA-18k.	Apache-2.0 License.
Qwen3-TTS [27]	Emotion-styled speech synthesis for dataset construction.	Apache-2.0 License.
Qwen3-Omni [18]	Spoken-dialogue baseline and audio-capable automated judging.	Apache-2.0 License.
Qwen2.5-Omni [48]	Spoken-dialogue baseline.	Apache-2.0 License.
Kimi-Audio [19]	Spoken-dialogue baseline.	MIT License for original code and Apache-2.0 License for Qwen-derived components, following the model card.
OpenS2S [21] and OSUM-EChat [23]	Spoken-dialogue baselines.	Apache-2.0 License.
VoiceBench CommonEval [35]	Real neutral spoken-query evaluation split.	Apache-2.0 License for VoiceBench; underlying Common Voice recordings are distributed under CC0-1.0 with Mozilla Common Voice terms.
AffectNet+ [13]	Facial expression sensing evaluation.	AffectNet/AffectNet+ academic research access terms; used only for non-commercial research evaluation.
SEED-VII [29]	EEG and eye-tracking sensing evaluation.	SEED dataset license agreement and application-based research access terms.
YAAD [30]	ECG and GSR sensing evaluation.	CC BY 4.0 License through Mendeley Data.
ISEAR [32]	Textual affect-description sensing evaluation.	Original research dataset credited to Scherer and Wallbott; used for research evaluation following the available public distribution terms.
HSEmotion [28]	Facial expression classifier implementation.	Apache-2.0 License.
Emotion English DistilRoBERTa-base [33] and all-MiniLM-L6-v2 [45, 46]	Text emotion classifier and query deduplication embeddings.	Hugging Face model-card terms; all-MiniLM-L6-v2 is Apache-2.0.
UTMOS [40], BERTScore [37], ROUGE-L [38], LoRA [25], and DeepSpeed [41]	Evaluation metrics and training infrastructure.	UTMOS and BERTScore are MIT License; ROUGE-score, LoRA/PEFT, and DeepSpeed are Apache-2.0 License.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state the paper’s main contributions: SYMPATHEIA, the SYMPATHEIA-18k dataset, continuous VA affect conditioning, and multimodal sensing integration. These claims are supported by the methodology, dataset construction, empirical results, and limitations sections.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: The paper includes a dedicated Limitations section discussing synthetic training data, reliance on automated audio-LLM evaluation, limits of single-point VA representations, noisy and privacy-sensitive external sensing, and computational cost of emotion-conditioned speech evaluation.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper does not present formal theoretical results, theorems, lemmas, or proofs. The mathematical expressions define the training objective and modality-to-VA mapping used in the empirical system, so there are no theoretical assumptions or proof obligations for this item.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper discloses the model architecture and conditioning mechanism in Section 3.2, dataset construction in Section 3.3, and implementation, training, VA-anchor, dataset-generation, evaluation, and sensing-module details in Appendix B. Anonymous artifacts provide code for fine-tuning, dataset generation, evaluation, and sensing-module experiments, and the full SYMPATHEIA-18k dataset is anonymously available at <https://huggingface.co/datasets/anonymous2222/Sympatheia-18k>.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.

- (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Project code for model fine-tuning, dataset generation, evaluation, and sensing-module experiments is anonymously available at <https://anonymous.4open.science/r/sympatheia-1181>. The full SYMPATHEIA-18k dataset is anonymously available at <https://huggingface.co/datasets/anonymous2222/Sympatheia-18k>, with implementation and experimental details provided in Appendix B.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The paper specifies dataset construction and splits in Section 3.3 and Appendix B.4, main fine-tuning details in Appendix B.2, and evaluation protocols in Sections 4.1–4.4 and Appendices B.6, B.7.1, B.7.2, and B.8. These include the 70/30 train/evaluation split, LoRA configuration, AdamW optimizer, learning rate, weight decay, batch size, gradient accumulation, warmup, number of epochs, gradient clipping, precision, DeepSpeed configuration, evaluation sampling, judge prompts, human-evaluation protocol, human facial-expression recording protocol, and sensing-module settings.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.

- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Appendix C.3 reports mean $\pm$ standard deviation for the primary evaluation metrics in Table 1, with standard deviations computed over evaluation examples for automated judge and similarity metrics and over human ratings for Emotion MOS. It also reports statistical significance annotations for the Spearman prosody correlations in Table 2, where bold coefficients indicate two-sided Spearman tests with $p < 0.01$.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B.2 reports the main fine-tuning setup, including precision, distributed training configuration, batch size, and training schedule. Appendix B.9 reports that experiments were run on a local compute cluster and gives the GPU type, GPU memory, and approximate runtimes for model fine-tuning, dataset construction, TTS synthesis, response generation, and automated judging.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The authors reviewed the NeurIPS Code of Ethics and the research conforms to its requirements. The paper reports human-study protocols in Appendix B.7, including participant recruitment and compensation for the Prolific human evaluation and volunteer participation, consent, and recording procedures for the human facial-expression study, and discusses privacy, consent, fairness, misuse risks, release safeguards, and out-of-scope uses in Appendix A.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Limitations section briefly flags deployment concerns and points to Appendix A, which discusses potential benefits for supportive and accessible voice interaction, risks related to over-trust, manipulation, privacy, consent, surveillance, and unequal performance across users and sensing conditions, and corresponding deployment safeguards.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: Appendix A describes safeguards for the model release, including documentation of intended research use, limitations, and out-of-scope uses. It also recommends opt-in sensing, disclosure, and user control for deployments that use external affect signals.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Existing models, datasets, and software assets are cited where they are introduced, and Appendix D lists the main external assets used in the paper together with their licenses or access terms. We used these assets for research, evaluation, or comparison purposes in accordance with their stated access conditions.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The paper introduces and releases the new SYMPATHEIA-18k dataset and accompanying code through anonymized artifact links in Appendix B.1. The dataset construction procedure, prompts, splits, valence–arousal anchors, training/evaluation details, limitations, and responsible-use considerations are documented in Sections 3.3 and Appendices B.4, B.5, B.3, B.2, and A. The released dataset page provides accompanying documentation for the asset, including its intended use, format, and limitations. The released dataset is synthetically generated; raw human-study recordings, including participant audio and video from the facial-expression recording study, are used only for evaluation and are not released.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [Yes]

Justification: Appendix B.7 describes the human Emotion MOS study, including Prolific recruitment, the number of participants, headphone/quiet-environment requirements, expected completion time, assignment of six-query batches, randomized anonymized model responses, the 1–5 emotional-appropriateness rating scale, and score aggregation. It also describes the end-to-end human facial-expression recording study, including the eight-subject within-subjects query–emotion design, volunteer participation, participant emotion-acting instructions, consent procedure, webcam/audio recording procedure, frame sampling, VA estimation, response-generation conditions, and non-release of raw recordings. Figure 4 shows the survey interface used by Emotion MOS participants, including the instructions and rating options. Emotion MOS participants were compensated through Prolific at a rate above the minimum wage in the authors’ country.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [Yes]

Justification: Appendix B.7 describes the Prolific human evaluation protocol, including participant recruitment, task structure, compensation, headphone/quiet-environment requirements, and anonymized response presentation, as well as the human facial-expression recording protocol. Participants were informed through study consent materials about the task, webcam/audio recording where applicable, compensation or volunteer status, data use, IRB information, and minimal potential risks such as fatigue or discomfort from listening to or acting emotion-related content. The end-to-end human facial-expression recording study collected webcam video and query audio for the described evaluation, stored recordings with coded subject identifiers, and does not release raw recordings. The study was conducted under an institutional IRB protocol covering human-subject psychophysics experiments; identifying institutional information is omitted for anonymous submission.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

1117 16. Declaration of LLM usage

1118 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1119 non-standard component of the core methods in this research? Note that if the LLM is used
1120 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1121 scientific rigor, or originality of the research, declaration is not required.

1122 Answer: [Yes]

1123 Justification: The paper documents all LLM and speech-LLM usage that is part of the
1124 research methodology: GLM-4-Voice-9B is used as the speech-to-speech backbone in
1125 Section 3.2; Qwen3-32B-Instruct and Qwen3-TTS are used for dataset construction in
1126 Section 3.3 and Appendix B.4; and Qwen3-Omni is used for audio-capable automated
1127 judging in Section 4.1 and Appendix B.6. Author-side use of LLM tools was limited to
1128 spell checking, grammar suggestions, editing assistance, and basic code assistance, which
1129 does not require separate declaration under the NeurIPS policy and did not affect the core
1130 methodology, scientific rigor, or originality of the work.

1131 Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.